Supplementary Material for "Observational Indistinguishability and Integrity Blind Regions in Hybrid Quantum-Classical Workflows"

Roberto Fernández-Barrios, Iker Pastor-López, Amaia Pikatza-Huerga, and Pablo García Bringas
Faculty of Engineering, University of Deusto, Bilbao, Spain

1 Purpose and Claim Boundary

This supplement records the complete formal statements with proofs, the experimental design, the adversary-model reading rule, the statistical units and their dependence structure, the validation checks, the full tables of the preregistered gates of artifacts 1.1.0, 1.2.0 and 1.3.0 (including the amendment that replaced the calibration rule and the adversarial cluster-preserving gate), the secondary model-impact profile and the reproduction mapping behind the main article. The experimental evidence is frozen at artifact 1.3.0; version 1.3.1 corrected the formal statements (state semantics, the three kernel notions, Proposition 7 under finite-shot estimation and the initial reference taxonomy) and editorial framing without re-executing any kernel, job or draw. Version 1.3.2 audits the actual sensor references, adopts repair B and adds only tables derived from frozen outputs; it changes no experimental result. Version 1.3.3 is a bibliographic/editorial correction only: no experiment, kernel, draw, model, seed, intervention, policy decision or scientific evidence was rerun or changed. It adds no QPU, calibrated-noise, scheduling, provider-security, multi-tenancy, deployed runtime-service or operational Fleet Management evidence. The exact claims remain bounded to data-to-evaluation interventions, ideal-statevector quantum kernels, controlled binomial shot emulation, offline policy evaluation on frozen outputs and a local executable contract.

2 Formal Core with Proofs

2.1 Objects

A workflow state consists of the primitive (stored) artifacts $s = (X, P, C, E, \xi, g, f, y) \in \mathcal{S}$ and of the derived artifacts the workflow recomputes from them: $\tilde{X} = P(X)$, the semantic kernel $K_{\text{sem}} = \Phi(C)$ on a fixed probe set, the finite-shot estimate $\hat{K} = \text{est}(K_{\text{sem}}, E, \xi)$ with estimation randomness ξ, the observed kernel $K_{\text{obs}} = g(\hat{K})$ delivered downstream after post-processing g, the predictions $\hat{y} = f(\tilde{X}, K_{\text{obs}})$ indexed by item identity $i = 1, \ldots, n$, and $R = r(\hat{y}, y)$. An intervention a overwrites a declared set of nodes (primitive or derived), keeps every node that is not a descendant of an overwritten node fixed, and recomputes the descendants of the overwritten nodes through the workflow with ξ retained; the baseline is s_0 and $s_a = a(s_0)$. Classes: T_y (label-only: overwrites y; only R is recomputed), $T_y^\pi \subset T_y$ (prior-preserving), the feature-side mechanisms (overwrite $\tilde{X}$ or X; $\hat{y}$ and R recomputed), including the adaptive cluster-preserving variants of Gate A, and, on the quantum branch, circuit-side (overwrite C), estimation-variation (overwrite ξ or E) and post-processing (overwrite g or K_{obs}) interventions. The executable counterpart of these semantics is `src/integrity/workflow_state.py`. The conclusion is $R(s)$ (balanced accuracy) and $\Delta_R(a) = R(s_0) - R(s_a)$; a is material at level τ if $|\Delta_R(a)| \geq \tau$. The limit $\tau \to 0^+$ is a structural-sensitivity endpoint (any conclusion change), not an operational risk threshold; $\tau = 0.02$ and 0.05 are the fixed larger-effect sensitivity analyses. An information set $\mathcal{I}$ is a view $V_{\mathcal{I}} : \mathcal{S} \to \mathcal{O}_{\mathcal{I}}$; refinement $\mathcal{I} \sqsubseteq \mathcal{I}'$ means $V_{\mathcal{I}} = \pi \circ V_{\mathcal{I}'}$; the join $\mathcal{I} \vee \mathcal{W}$ is the pair of views. A reference is stored information about the baseline that the auditor compares the audited batch with; it is trusted when no intervention of the class can alter it. We describe references on three separate axes: provenance (historical, benchmark-protected, or deployed authenticated), granularity (aggregate or item-aligned), and decision (statistical or exact). The compact classes are not a one-dimensional trust scale. Class/Level A denotes statistically thresholded aggregate comparison and may use a historical population or a clean version of the same item set. The executed A sensors use the latter as a benchmark-protected oracle without item-wise pairing; no deployed authentication mechanism is demonstrated. Level B is a deployed-trusted aggregate same-batch reference $\rho = V_{\mathcal{J}}(s_0)$ with $V_{\mathcal{J}}$ permutation-invariant; Level C is a deployed-trusted item-aligned same-batch reference with $V_{\mathcal{J}}$ item-indexed. Thus same items in a benchmark computation does not imply a trusted deployed anchor. Every selected batch sensor in the executed policy belongs to Class A; the sensor-level audit below records the exact current and reference values.

2.2 Statements and Proofs

Lemma 1 (structural blindness). If $V_{\mathcal{I}}(s_a) = V_{\mathcal{I}}(s_0)$ then $S(s_a) = S(s_0)$ for every sensor S admissible in $\mathcal{I}$.

Proof: $S(s_a) = S(V_{\mathcal{I}}(s_a)) = S(V_{\mathcal{I}}(s_0)) = S(s_0)$. For a randomized sensor whose seed is part of the view the identity holds path-wise; with a fresh seed the two outputs are equal in distribution. □

Definition (blind regions and gaps). $B_{\mathcal{I}}(\mathcal{A}) = \{a \in \mathcal{A} : V_{\mathcal{I}}(a(s_0)) = V_{\mathcal{I}}(s_0)\}$; $B_{\mathcal{I},\mathcal{S}}(\mathcal{A}) = \{a : S(a(s_0)) = S(s_0)\ \forall S \in \mathcal{S}\}$; $G_{\mathcal{I}}(\tau) = \{a : |\Delta_R(a)| \geq \tau, a \in B_{\mathcal{I}}\}$ and $G_{\mathcal{I},\mathcal{S}}(\tau)$ analogously. By Lemma 1, $B_{\mathcal{I}} \subseteq B_{\mathcal{I},\mathcal{S}}$ and $G_{\mathcal{I}} \subseteq G_{\mathcal{I},\mathcal{S}}$, with equality iff $\mathcal{S}$ is baseline-separating on the orbit of $\mathcal{A}$: it takes a different value at $a(s_0)$ than at s_0 whenever the views differ. Pairwise separation of intervened states (injectivity on the orbit), which identification of the intervention would need, is never assumed.

Proposition 1 (monotonicity). If $\mathcal{I} \sqsubseteq \mathcal{I}'$ then $B_{\mathcal{I}'}(\mathcal{A}) \subseteq B_{\mathcal{I}}(\mathcal{A})$ and $G_{\mathcal{I}'}(\tau) \subseteq G_{\mathcal{I}}(\tau)$.

Proof: If $V_{\mathcal{I}'}(s_a) = V_{\mathcal{I}'}(s_0)$ then $V_{\mathcal{I}}(s_a) = \pi(V_{\mathcal{I}'}(s_a)) = \pi(V_{\mathcal{I}'}(s_0)) = V_{\mathcal{I}}(s_0)$. Membership in the gap adds only the materiality condition, which does not depend on the view. □

Corollary 1. With f fixed and deterministic: (a) $T_y \subseteq B_{\mathcal{I}_{XF}} \subseteq B_{\mathcal{I}_X}$ and every sensor admissible in $\mathcal{I}_X$ or $\mathcal{I}_{XF}$ is invariant under a label-only change; (b) $T_y^{\pi} \subseteq B_{\mathcal{I}_{Y_m}}$ and every sensor admissible in $\mathcal{I}_{Y_m}$ is invariant under a prior-preserving change.

Proof: (a) T_y fixes X, P and f, hence $\tilde{X}$ and $\hat{y} = f(\tilde{X})$; the multiset of pairs $(\tilde{x}_i, \hat{y}_i)$ is unchanged. Apply Lemma 1 and Proposition 1 with $\mathcal{I}_X \sqsubseteq \mathcal{I}_{XF}$. (b) T_y^{π} preserves the class histogram by definition. □

Proposition 2 (closure). $B_{\mathcal{I}\vee\mathcal{W}}(\mathcal{A}) = B_{\mathcal{I}}(\mathcal{A}) \cap B_{\mathcal{W}}(\mathcal{A})$; a class $\mathcal{A} \subseteq B_{\mathcal{I}}$ is completely separable in $\mathcal{I} \vee \mathcal{W}$ iff $V_{\mathcal{W}}(a(s_0)) \neq V_{\mathcal{W}}(s_0)$ for every $a \in \mathcal{A}$.

Proof: The joint view equals the baseline joint view iff both components do. Necessity: an a that leaves $V_{\mathcal{W}}$ unchanged stays in the joint blind region. Sufficiency: a changed component changes the pair. □

Corollary 2. With f fixed and deterministic, every view factoring through $(\tilde{X}, f(\tilde{X}), \mathrm{hist}(y))$ leaves every $a \in T_y^{\pi}$ in its blind region. The multiset $\mathcal{I}_{XFY}$ separates $a \in T_y^{\pi}$ unless a permutes labels only among items with identical $(\tilde{x}, \hat{y})$; an item-aligned reference $\mathcal{I}^{\star}_{XFY}$ separates every non-identity relabeling.

Proof: Apply Proposition 2 with $\mathcal{W}$ the view in question: by Corollary 1 each of the three components is invariant, so their join is. For the multiset of triples, two states have the same multiset iff a permutation of items maps one to the other; since $(\tilde{x}_i, \hat{y}_i)$ is fixed, such a permutation can only move labels among items with identical $(\tilde{x}, \hat{y})$. An item-indexed tuple has no such freedom. □

Proposition 3 (materiality forces aggregate separability). If $R = g(M)$ with M the confusion matrix, $a \in T_y$ and $\Delta_R(a) \neq 0$, then $M(s_a) \neq M(s_0)$; hence a is separable in the confusion view and in $\mathcal{I}_{XFY}$, and $G_{\mathcal{I}_{XFY}}(\tau) \cap T_y = \emptyset$ for $\tau > 0$.

Proof: $M(s_a) = M(s_0)$ implies $R(s_a) = g(M(s_a)) = g(M(s_0)) = R(s_0)$. The confusion matrix is a function of the multiset of triples, so a changed M implies a changed multiset. □

Remark (other metrics). The proof uses only $R = g(M)$, so the statement holds for accuracy, precision, recall, F_1 and every function of the binarised confusion matrix, with M as the sufficient aggregate. ROC-AUC is a function of the multiset of (score, label) pairs and not of M: two label vectors with the same confusion matrix can have different AUC. The same argument then holds with that multiset as the aggregate: a label-only change that changes the AUC changes the score-label multiset, a trusted stored copy of it detects every material change exactly, and item identity again needs an item-aligned reference. Both facts are checked by enumeration in the tests of the artifact; no AUC experiment is run.

Proposition 4 (separability and anchoring). (i) For any auditor whose decision is a function of $V_{\mathcal{I}}(s)$ (including a seed carried in the view), $a \in B_{\mathcal{I}}$ implies $\mathrm{fire}(s_a) = \mathrm{fire}(s_0)$. (ii) A reference-anchored auditor with trusted $\rho = V_{\mathcal{J}}(s_0)$ (level B or C) and rule "fire iff $d(V_{\mathcal{J}}(s), \rho) > 0$" has zero false-alarm probability and detects every $a \notin B_{\mathcal{J}}$ with probability one.

Proof: (i) is Lemma 1 applied to the decision function. (ii) $d(\rho, \rho) = 0$ and $d(V_{\mathcal{J}}(s_a), \rho) > 0$ iff $V_{\mathcal{J}}(s_a) \neq \rho$, which is $a \notin B_{\mathcal{J}}$ because ρ is unaltered. □

Corollary 3 (which reference certifies which integrity). Let $a \in T_y$ with f fixed and deterministic. (a) With a trusted aggregate reference $M_0 = M(s_0)$ of the same batch, "fire iff $M(s_a) \neq M_0$" detects every violation of aggregate integrity and, by Proposition 3, every material a; a trusted R_0 detects every material a and nothing else. (b) Any reference that factors through M, $\mathrm{hist}(y)$ or R is invariant under relabelings that permute labels among items with equal predictions (counterexample C1), which violate item-identity integrity; an item-aligned reference y_0 detects every non-identity relabeling.

Proof: (a) Proposition 4(ii) with $\mathcal{J}$ the confusion view, then Proposition 3. (b) Swapping y_i, y_j with $\hat{y}_i = \hat{y}_j$ moves one item from cell $(\hat{y}_i, y_i)$ to $(\hat{y}_i, y_j)$ and another from $(\hat{y}_j, y_j)$ to $(\hat{y}_j, y_i)$; the moves cancel in M, hence in every function of M, and the histogram is unchanged, whereas the item-indexed tuple differs in positions i and j. □

Proposition 5 (union of calibrated sensors versus conformal family calibration). (a) Let $E_1, \ldots, E_m$ be the null firing events of m sensors with $p_j = \Pr_0[E_j] \leq \alpha$. Then

$$\max_j p_j \ \leq\ \Pr_0\Big[\bigcup_j E_j\Big] \ \leq\ \min\Big(1, \sum_j p_j\Big) \ \leq\ \min(1, m\alpha). \tag{1}$$

Without assuming that some $p_j = \alpha$ the universal lower bound is $\max_j p_j$, which can be 0. If every $p_j = \alpha$, then $\alpha \leq \Pr_0[\bigcup_j E_j] \leq \min(1, m\alpha)$: the lower bound is attained by nested (coincident) events, the upper bound $m\alpha$ (when $m\alpha \leq 1$) by mutually disjoint events, and under independence $\Pr_0[\bigcup_j E_j] = 1 - \prod_j (1 - p_j) = 1 - (1-\alpha)^m$, strictly between the two bounds for $m \geq 2$ and $0 < \alpha < 1$.

(b) Let $v_1, \ldots, v_n \in \mathbb{R}^m$ be the sensor vectors of the calibration draws and v_{n+1} that of the audited batch. Give every member j of the augmented set the leave-one-out family score

$$\tilde{U}_j = \max_{s \leq m} \frac{\#\{i \neq j : v_{s,i} < v_{s,j}\}}{n}, \tag{2}$$

computed against the other n members, and define $p = (1 + \#\{i \leq n : \tilde{U}_i \geq \tilde{U}_{n+1}\})/(n+1)$; the family fires iff $p \leq \alpha$, ties counting against firing. Let $k = \lfloor \alpha(n+1) \rfloor$ and $N_j = \#\{i \leq n+1 : \tilde{U}_i \geq \tilde{U}_j\}$. Then (i) for every augmented set at most k members satisfy $N_j \le k$; (ii) if the $n+1$ vectors are exchangeable,

$$\Pr_0[\, p \le \alpha \,] \;\le\; \frac{\lfloor \alpha(n+1) \rfloor}{n+1} \;\le\; \alpha, \tag{3}$$

with equality in the first inequality when the scores are almost surely distinct, and with no assumption on ties or continuity; (iii) for a single sensor without ties the rule coincides with "fire iff v_{n+1} exceeds the $\lceil (n+1)(1-\alpha) \rceil$-th smallest calibration value"; (iv) the level is marginal over the joint draw of calibration set and audited batch, audited batches sharing one calibration set are dependent, and nothing is claimed when exchangeability fails. For $n = 200$, $\alpha = 0.05$: $k = 10$ and the level is $10/201 = 0.0498$.

(c) The rule of artifact 1.2.0 scored calibration draws against the other $n-1$ calibration draws only, the audited batch against all n, and fired when the audited score exceeded the $\lceil (n+1)(1-\alpha) \rceil$-th smallest calibration score; it claimed the level $\alpha + 1/(n+1)$. The claim is false. For $n = 4$, $\alpha = 0.2$ and three sensors with cyclic orderings $s_1 : j_1 > j_2 > j_3 > x > y$, $s_2 : j_2 > j_3 > j_1 > x > y$, $s_3 : j_3 > j_1 > j_2 > x > y$, the members j_1, j_2, j_3 each fire when audited (calibration scores $0.75, 0.75, 0.25, 0$, threshold 0.75, audited score 1), so the firing probability under exchangeability is 0.60 against the claimed 0.40 and against α. On the frozen null draws the rule exceeds α under exchangeable re-splits (Table 16).

Proof: (a) The union contains each E_j (lower bound); Boole's inequality gives the upper bound; nested, disjoint and independent events attain or realise the stated values, the last by inclusion–exclusion. (b)(i) Suppose $k+1$ members satisfy $N_j \le k$ and let j^* be one with the smallest score among them; every one of the $k+1$ has $\tilde{U} \ge \tilde{U}_{j^*}$, so $N_{j^*} \ge k+1$, a contradiction. (ii) The score of member j depends on v_j and on the multiset of the others, so the score map is permutation-equivariant and exchangeability of the v's gives exchangeability of the $\tilde{U}$'s and of the N_j's; $p \le \alpha$ iff $N_{n+1} \le k$, and by exchangeability $\Pr[N_{n+1} \le k] = \mathbb{E}[\#\{j : N_j \le k\}]/(n+1) \le k/(n+1)$ by (i). With distinct scores exactly k members satisfy $N_j \le k$; ties merge ranks and shrink that set. (iii) For $m = 1$ and distinct values, $\tilde{U}_i \ge \tilde{U}_{n+1}$ iff $v_i > v_{n+1}$, so $p = (1 + \#\{i : v_i > v_{n+1}\})/(n+1) \le \alpha$ iff v_{n+1} exceeds at least $n - k + 1 = \lceil (n+1)(1-\alpha) \rceil$ calibration values. (iv) restates what (ii) proves. (c) is checked by enumeration in the artifact tests, which also verify (i) over every weak ordering (all tie patterns) of up to seven elements with one sensor, five with two sensors and four with three sensors, and over random tied configurations with up to ten sensors. □

Proposition 6 (no local authentication without an uncontrolled root). Let a *local verifier* be any map $\mathrm{acc} : \mathcal{O}_{\mathcal{I}} \times \mathrm{Ref} \to \{\text{accept}, \text{reject}\}$ of the observed view and a reference bundle, let $\mathcal{S}_H$ be the honest family of states and $\rho_H(s) = V_{\mathcal{J}}(s)$ the reference the honest process attaches to s. Assume (honest completeness) $\mathrm{acc}(V_{\mathcal{I}}(s), \rho_H(s)) = \text{accept}$ for every $s \in \mathcal{S}_H$, and (joint control) the adversarial class can produce $(V_{\mathcal{I}}(s'), \rho_H(s'))$ for some $s' \in \mathcal{S}_H$ with $s' \not\sim_{\mathcal{I}} s_0$. Then the verifier accepts the substituted pair, which is observationally an honest run of s', so no function of that input establishes authenticity of the served state relative to s_0; if $R(s') \ne R(s_0)$ a material change is served. Conversely, if a component $\rho^\star = V_{\mathcal{K}}(s_0)$ of the bundle cannot be written by the class, "reject iff $V_{\mathcal{K}}(s) \ne \rho^\star$" rejects every substitution with $V_{\mathcal{K}}(s') \ne V_{\mathcal{K}}(s_0)$.

Proof: The substituted input equals the honest input of s'; by honest completeness it is accepted, and acceptance is the same event as for an honest run of s', so no function of the input distinguishes the two. The converse is Proposition 4(ii) applied to $\mathcal{K}$. The statement concerns authenticity, not consistency: by Proposition 3 a material substitution changes the joint view, but the change is invisible to a verifier all of whose evidence the class can rewrite. □

Proposition 7 (the quantum branch as an instance of the lattice). Let $\mathcal{C}$ be the circuit representations, $\Phi : \mathcal{C} \to \mathcal{K}$ the semantic kernel map on a fixed probe set, $K_{\mathrm{sem}} = \Phi(C)$, $\hat{K}$ the finite-shot estimate, K_{obs} the observed kernel, h the provenance hash, A the algebraic invariants and f_K the decision from a kernel. Three intervention classes are distinguished: (a) circuit-side (overwrites C; may change K_{sem}); (b) estimation variation (C and K_{sem} fixed; $\hat{K}$ changes); (c) post-processing or kernel substitution (C, K_{sem} and $\hat{K}$ fixed; K_{obs} changes). Assume h collision-free on the circuits under study. (i) On class (a), $[C] = \Phi^{-1}(\Phi(C))$ is the semantic class of C; approved rewrites map C into $[C]$ without fixing $h(C)$, so V_h refines $V_{K_{\mathrm{sem}}}$ and $B_h(\mathcal{A}) \subseteq B_{K_{\mathrm{sem}}}(\mathcal{A})$, strictly whenever $\mathcal{A}$ contains an approved rewrite. (ii) On class (c), $V_A = A \circ V_{K_{\mathrm{obs}}}$ and $V_f = f.(\tilde{X}) \circ V_{K_{\mathrm{obs}}}$, so $B_{K_{\mathrm{obs}}} \subseteq B_A$ and $B_{K_{\mathrm{obs}}} \subseteq B_f$; a PSD-preserving substitution $K' \ne K_{\mathrm{obs},0}$ with $A(K') = A(K_{\mathrm{obs},0})$ lies in $B_A \setminus B_{K_{\mathrm{obs}}}$ and, because C and K_{sem} are fixed on this class, also in $B_h \cap B_{K_{\mathrm{sem}}}$: it is closed only by a trusted reference on K_{obs} itself, so the trusted semantic probe of C and the trusted observed-kernel reference are different anchors. A kernel change that crosses no decision boundary lies in $B_f \setminus B_{K_{\mathrm{obs}}}$. (iii) On class (b), for an honest finite-shot estimate the acceptance rule "accept iff $\hat{K} = K_{\mathrm{sem},0}$" has false-alarm probability $1 - \Pr[\hat{K} = K_{\mathrm{sem},0} \mid \text{honest}]$, which depends on the discrete support of the estimator and may be large or equal to one but is not universally one; exact equality is therefore not an appropriate acceptance criterion under finite-shot estimation, $d(\hat{K}, K_{\mathrm{sem},0})$ must be treated as a level-A statistic with a null from repeated honest estimation, to which Proposition 5(b) applies with $m = 1$ when the repeated estimates and the audited one are exchangeable, and where no such null is calibrated no statistical integrity claim is made. No inclusion is claimed across classes.

Proof: (i) If $h(C_1) = h(C_2)$ then $C_1 = C_2$ by collision freedom and $\Phi(C_1) = \Phi(C_2)$; hence V_h refines $V_{K_{\mathrm{sem}}}$ and Proposition 1 gives the inclusion; an approved rewrite $C' \in [C_0]$ with a different canonical text has $h(C') \ne h(C_0)$ and $\Phi(C') = \Phi(C_0)$. (ii) is Proposition 1 applied to the two coarsenings of $V_{K_{\mathrm{obs}}}$; on class (c) the primitive C is untouched, so $h(C)$ and $\Phi(C)$ are unchanged by the definition of an intervention (Section 2.1), which gives the membership in $B_h \cap B_{K_{\mathrm{sem}}}$. The set differences are witnessed in Table 6 (15/15 PSD-preserving substitutions with silent algebraic sensors and unchanged circuit hash, exposed by the anchored comparison of the delivered kernel; RZ mutations at 0.02 with zero output change and firing semantic sensor). (iii) For an entry with fidelity p estimated from N shots, $N\hat{p} \sim \mathrm{Binomial}(N, p)$ and $\Pr[\hat{p} = p] = \binom{N}{Np} p^{Np} (1-p)^{N-Np}$

TABLE 1
Counterexample witnesses realised in the frozen expansion (count / denominator). W1–W4 are evaluation-label rows, W5–W6 feature-side rows, W7 all intervened rows, W8 material label rows; W9 and W10 count the label rows that a trusted aggregate reference, respectively only an item-aligned reference, detects (Corollary 3).

ID	Property	Count
W1	identity change without joint-outcome or conclusion change	808 / 3,600
W2	conclusion change without model-output change	2,617 / 3,600
W3	marginal invariance with conclusion change	1,059 / 1,800
W4	confusion-profile change without conclusion change	175 / 3,600
W5	model-output change without conclusion change	296 / 7,200
W6	representation change without output change	2,513 / 7,200
W7	apparent improvement under intervention	1,534 / 10,800
W8	material conclusion change always changes the confusion profile (label path)	2,617 / 2,617
W9	label rows detectable by a trusted aggregate (confusion-matrix) reference of the same batch	2,792 / 3,600
W10	label rows detectable only by an item-aligned reference	808 / 3,600

when Np is an integer and 0 otherwise; for $p = 1/2$ and $N = 2$ this is $1/2$, so the false-alarm probability of exact equality is $1/2$, not one, and for $p = 1/3$, $N = 2$ it is one. For a kernel of independently estimated entries the probabilities multiply. The universal statement "$\Pr[\hat{K} = K_{\mathrm{sem},0}] = 0$ under non-degenerate shot noise" of the earlier version is therefore false on the support of the estimator and is withdrawn; the conclusion that exact equality is not an acceptance criterion and that $d(\hat{K}, K_{\mathrm{sem},0})$ needs a calibrated null stands, with the rest being the definition of the batch-level auditor and Proposition 5(b). The arithmetic is checked in `tests/test_workflow_state.py`. □

2.3 Counterexamples

Table 1 restates the witnesses of the main text. C1 is the swap of two labels between items with equal predictions; C3 between items with different predictions; C4 uses two compensating flips that move the same number of items across the diagonal of M in each class; C5 and C6 are feature-side; C7 counts every intervened row whose balanced accuracy increased; C8 is Proposition 3; W9 and W10 count the label rows that a trusted aggregate reference of the same batch detects (every material one) and those that only an item-aligned reference exposes (Corollary 3). The cluster-preserving drift of Gate A (C9 in the artifact's formal core) is not a structural blind region but a concrete stress test of sensor insufficiency against an adaptive attacker. It is measured in Section 12, not proved or presented as priority for generic monitor-aware drift evasion.

3 Adversary Model: Reading Rule

Every executed intervention is assigned to one of four classes: *fault robustness* (benign or accidental change), *integrity corruption* (non-adaptive change by a party that does not model the auditor), *adversarial attack* (targeted change to alter the reported conclusion) and *adaptive attacker* (targeted change designed with knowledge of the monitored statistics). The complete per-class record (actor, capability, access point, knowledge, objective, budget, alterable assets, roots the class cannot write, observable information set and supported claim) is `manuscript/ADVERSARY_MODEL.md` in the artifact; the main article prints the summary table. The trusted computing base is: the local runtime and host; the training data, training routine and fitted model; the reference input array and preprocessing declaration; the reference circuit and probe kernels; SHA-256 collision resistance; the item-level label reference only where $\mathcal{I}^{\star}_{XFY}$ is declared; and the code that builds and verifies the envelope.

4 Frozen Experimental Design

4.1 Expansion Environments

The expansion comprises eight fixed environments (Table 2): five from CICIDS2017, two from UNSW-NB15 and one from ToN-IoT; with Gate 1, six of the nine environments of the study are CICIDS2017. "OOD" means a prespecified source-to-target construction, not a sampled deployment population. Every environment uses five split seeds and two nested model seeds and evaluates projected dimensions 8, 10 and 12. The CICIDS scale gate uses 256/256 train/evaluation samples; all others use 128/128.

Each staged table contains 3,000 balanced binary rows before experimental subsampling. Numeric non-finite values are replaced during staging; categorical fields are excluded. Several temporal targets yield clean performance near chance, limiting the impact that later interventions can express; the exact information-set invariances do not depend on clean headroom.

4.2 Perturbation Suite

The paper-core suite crosses six mechanisms with nominal strengths 0.02, 0.05 and 0.10 (Table 3). Artifact identifiers retain the historical family string `target_shift`; the scientific text uses *evaluation-label intervention* because these rows do not instantiate the statistical label-shift problem. Gate A adds cluster-preserving variants of mechanisms 4–9 at strengths 0.02, 0.05, 0.10, 0.25 and 0.50 as a concrete monitor-aware stress test of the declared fingerprint (Section 12).

5 Information-Set Coverage Contract

Table 5 records what each evidence regime can establish in the implemented study. Structural blindness means that the view is unchanged by construction; it is stronger and narrower than observing zero empirical response in a finite sample, and both are distinct from calibrated detectability and from adaptive evasion.

The implementation chain is explicit. The runner constructs feature, prediction, label-marginal and confusion-profile scores from the current batch and the clean benchmark-held arrays; the statistics consume aggregate distributions and not item correspondence. Calibration converts those scores to family flags. Gate D and `src/hsaas/policy.py` consume only the union flag, family flags and exact-reference flag—never raw `X_te_f`, `y_te_f`, or a clean batch. Producing the selected Class-A scores upstream nevertheless requires the benchmark oracle shown in Table 4. The benchmark protects that oracle from its executed attack API, but the artifact neither authenticates nor delivers it to a deployed auditor. By contrast, Class B/C exact flags presuppose a deployed-authenticated commitment at the declared granularity.

TABLE 2
Prespecified expansion environments.

ID	Environment	Training source	Evaluation source	Samples	Profiles
E1	CICIDS scale/ID	Frozen CICIDS subset	Same staged source	256/256	SVC, ZZ
E2	UNSW-NB15 ID	Balanced staged UNSW	Same staged source	128/128	SVC, ZZ, PauliXYZ, Z/PauliXZ-equivalent
E3	ToN-IoT ID	Balanced staged ToN-IoT	Same staged source	128/128	SVC, ZZ, PauliXYZ, Z/PauliXZ-equivalent
E4	CICIDS Tue→Wed	Tuesday traffic	Wednesday traffic	128/128	SVC, ZZ
E5	CICIDS Tue→Fri-PortScan	Tuesday traffic	Friday PortScan traffic	128/128	SVC, ZZ
E6	CICIDS Wed→Thu-Web	Wednesday traffic	Thursday Web traffic	128/128	SVC, ZZ
E7	CICIDS Wed→Fri-AM	Wednesday traffic	Friday morning traffic	128/128	SVC, ZZ
E8	UNSW temporal OOD	Prespecified UNSW source period	Prespecified UNSW target period	128/128	SVC, ZZ

TABLE 3
Frozen 18-condition intervention suite.

IDs	Mechanism	Strengths
1–3	Feature sign flip	0.02, 0.05, 0.10
4–6	Feature-wise mean shift	0.02, 0.05, 0.10
7–9	Scaling drift	0.02, 0.05, 0.10
10–12	Feature dropout	0.02, 0.05, 0.10
13–15	Prior-preserving evaluation-label flip	0.02, 0.05, 0.10
16–18	Random evaluation-label flip	0.02, 0.05, 0.10

Every machine-readable coverage record contains eight fields: workflow boundary; protected asset; adversary or failure class including fixed elements; available information regime and trusted references; sensor, calibration rule, its level and premise; covered perturbations and structural, calibrated and adaptive residual blind regions; cost on benign variation; response or countermeasure.

6 Deduplication and Statistical Units

Gate 1 emits 7,980 raw model-intervention rows; different quantum-map jobs repeat the same classical SVC pipeline, producing 3,420 repeated rows that are accepted as duplicates only after numerical identity checks, leaving 4,560 unique observations. The expansion emits 13,680 raw rows from 360 completed CSV/JSON job pairs with 2,280 verified repeats, leaving 11,400 unique observations. The strict builder fails if a job is missing, an environment contains an unexpected design, or a duplicated row disagrees.

Coverage counts use prespecified cells as denominators. For the secondary suite-average model profile, nested model-seed values are averaged inside each of five split seeds; the split seed is the inferential unit and the interval is $\bar{d} \pm t_{0.975,4}\, s_d/\sqrt{5}$. For the calibration gates the dependence structure is explicit: each of the 240 runs (eight environments $\times$ five split seeds $\times$ two model seeds $\times$ three dimensions) draws 20 calibration and 20 evaluation batches as overlapping simple random subsets of one pool half, the 200 draws of a cell come from ten pool halves, and the 200 evaluation draws of a cell share one calibration set; the inferential unit is the (environment, split-seed) cluster (40 clusters, five per environment). Pooled rates over draws and the Clopper–Pearson intervals of artifact 1.1.0 are descriptive. The conformal level of Proposition 5(b) is marginal, not conditional on the calibration set. No multiplicity adjustment or population transport is claimed.

7 Exact-Statevector Engine Validation

The expansion evaluator replaces pairwise compute-uncompute execution with a mathematically equivalent ideal-statevector calculation: it prepares each unique state $|\phi(x)\rangle$ once and forms $K(x, y) = |\langle\phi(x) \mid \phi(y)\rangle|^2$ by matrix multiplication, with the same eigenvalue-clipping PSD repair as the frozen Qiskit reference. Validation has four layers: symmetric and cross-kernel unit matrices agree with the reference at relative and absolute tolerance 10^{-10}; symmetry, unit diagonal, numerical range, cache bounds and cache reuse are unit-tested; a CICIDS 128/128 end-to-end replay (split/model seed 42, dimension 8, ZZ, paper-core suite) preserves every categorical, discrete performance, impact and integrity-signal endpoint, with only ROC-AUC and its derived drops differing by at most 1.221896×10^{-4} (one half-tie increment for 64 positive and 64 negative examples); and the prespecified 256/256 pilot completes. The 240 jobs of Gate A add a fifth layer: their clean rows and matched controls reproduce the frozen expansion rows with a maximum absolute difference of 0.0 in thirteen columns over 4,200 rows. The engine is tagged `exact_statevector`; it supplies no finite-shot, noise-model or hardware evidence.

8 Quantum Gate: Conditions and Coverage

The quantum gate crosses five split seeds, dimensions 4/6/8 and 11 conditions, giving 165 cells. Table 6 and Fig. 1 report response fractions over the 15 split-dimension cells per condition. A value of one means non-zero raw response in all 15 cells at the frozen tolerance; it is not calibrated power. The table is the empirical realisation of Proposition 7, read class by class: rows 2–3 (class (a)) change the hash and nothing semantic (i); rows 7–9 (class (c)) change the observed kernel with the circuit hash silent and are or are not visible to the algebraic coarsening (ii); rows 10–11 (class (b)) show a non-zero repeated-estimation discrepancy in every one of these cells, an empirical property of the gate that

TABLE 4
Audited reference semantics of every selected sensor in v1.3.2. Pairing means that item correspondence enters the statistic; same set means the runner uses the clean version of the audited item set. Harness protected means that the benchmark keeps the reference outside the executed attack API; trusted means that the threat model grants deployed authentication. Hist./stat. reports historical and statistical reference semantics in that order. Class A is statistical comparison; B and C are trusted aggregate and trusted item-aligned exact invariants.

Sensor	Current value	Reference value	Item pair	Same set	Harness protected	Threat-model trusted	Hist. / stat.	Rule / class
`integrity_jsd_vs_clean_eval`	empirical feature distributions of X_eval	X_te_f clean feature distributions	no	yes	yes	no	no / yes	statistical / A
`integrity_mmd_vs_clean_eval`	feature multiset X_eval	feature multiset X_te_f	no	yes	yes	no	no / yes	statistical / A
`integrity_ks_reject05_vs_clean_eval`	per-feature distributions of X_eval	per-feature distributions of X_te_f	no	yes	yes	no	no / yes	statistical / A
`integrity_score_jsd_vs_clean_eval`	score distribution f(X_eval)	score distribution f(X_te_f)	no	yes	yes	no	no / yes	statistical / A
`integrity_pred_pos_rate_shift`	mean predicted-positive rate on X_eval	mean predicted-positive rate on X_te_f	no	yes	yes	no	no / yes	statistical / A
`integrity_pred_jsd`	predicted-class distribution on X_eval	predicted-class distribution on X_te_f	no	yes	yes	no	no / yes	statistical / A
`integrity_label_prior_shift`	mean positive-label rate of y_eval	mean positive-label rate of y_te_f	no	yes	yes	no	no / yes	statistical / A
`integrity_label_jsd`	binary label distribution of y_eval	binary label distribution of y_te_f	no	yes	yes	no	no / yes	statistical / A
`integrity_confusion_profile_l1`	aggregate confusion PMF of (y_eval,f(X_eval))	aggregate confusion PMF of (y_te_f,f(X_te_f))	no	yes	yes	no	no / yes	statistical / A
`integrity_confusion_profile_jsd`	aggregate confusion PMF of (y_eval,f(X_eval))	aggregate confusion PMF of (y_te_f,f(X_te_f))	no	yes	yes	no	no / yes	statistical / A
`integrity_confusion_profile_l1_delta`	aggregate confusion-profile L1 delta	trusted aggregate confusion profile of the same batch	no	yes	yes	yes	no / no	exact / B
`integrity_confusion_profile_jsd_delta`	aggregate confusion-profile JSD delta	trusted aggregate confusion profile of the same batch	no	yes	yes	yes	no / no	exact / B
`integrity_pred_disagreement`	item-indexed predictions f(X_eval)	trusted item-indexed predictions f(X_te_f)	yes	yes	yes	yes	no / no	exact / C
`label_flip_rate`	item-indexed labels y_eval	trusted item-indexed labels y_te_f	yes	yes	yes	yes	no / no	exact / C

TABLE 5
Evidence regimes, implemented sensors, demonstrated blind regions and calibrated status (conformal family rule).

Regime	Evidence	Implemented sensor family	Structural blind region	Calibrated / exact status
$\mathcal{I}_X$	Features	Feature JSD, MMD, KS rejection	Every label-only change	Conformal decision FPR 0.056; label path 0/3,600; evaded by cluster-preserving drift at matched strengths
$\mathcal{I}_{XF}$	Features, scores, predictions	$\mathcal{I}_X$ plus score JSD, prediction-rate shift, prediction JSD	Every label-only change for a fixed predictor	Conformal FPR 0.058; label path 0/3,600; partially evaded by cluster-preserving drift
$\mathcal{I}_{Y_m}$	Label class counts	Prior delta and label JSD	Prior-preserving relabeling; every feature-side change	FPR 0.048; zero containment; 5,450 residual-blind material cases at $\tau \to 0^+$
$\mathcal{I}_{XFY}$ (batch)	Item triples; Class-A statistical comparison	Confusion-profile statistics added	None for material label changes (Proposition 3) but statistically undetectable: 11–43 of 2,617	Conformal FPR 0.053
$\mathcal{I}^{\star}_{XFY}$	Deployed-trusted same-batch references (B aggregate; C item-aligned)	Aggregate confusion-profile deltas; item-aligned prediction disagreement and label mismatch (exact-zero null)	None	Exact: 2,617/2,617 material label rows and every material adaptive row; 0 material corruptions served; 0 invariant violations on 1,200 exact-zero rows; 85 gross exact blocks and 39 net additional near-null interruptions versus batch $\mathcal{I}_{XFY}$/P2
$\mathcal{I}_Q$	Circuit, parameters, kernel, estimation evidence	Hashes, semantic probes, algebra, repeated estimation, output changes	Composition-dependent: hashes overreact to equivalence; algebra misses PSD-preserving substitution; outputs miss sub-decision changes (Proposition 7)	165/165 cells; nine acceptance checks

TABLE 6
Quantum-gate sensor response fractions.

Condition	Circuit prov.	Kernel prov.	Semantic kernel	Algebraic kernel	Repeated estimate	Output change
Clean	0.00	0.00	0.00	0.00	0.00	0.00
Benign transpilation	1.00	1.00	0.00	0.00	0.00	0.00
Common-unitary rewrite	1.00	1.00	0.00	0.00	0.00	0.00
Parameterized RZ 0.02	1.00	1.00	1.00	0.00	0.00	0.00
Parameterized RZ 0.10	1.00	1.00	1.00	0.00	0.00	0.13
Feature-map reps 1→2	1.00	1.00	1.00	0.00	0.00	0.73
Asymmetric kernel edit	0.00	1.00	1.00	1.00	0.00	0.00
Kernel diagonal erosion	0.00	1.00	1.00	1.00	0.00	0.00
PSD-preserving kernel mix	0.00	1.00	1.00	0.00	0.00	0.07
Shot estimator 256	0.00	1.00	1.00	1.00	1.00	0.47
Shot estimator 1,024	0.00	1.00	1.00	1.00	1.00	0.20

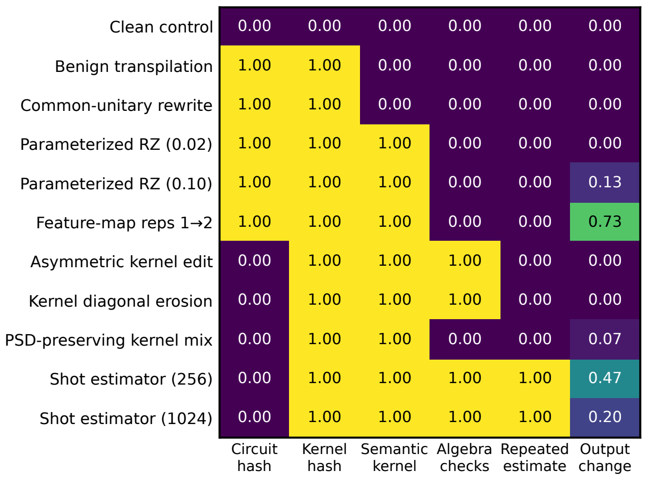


Fig. 1. Response fraction of each quantum-branch sensor over the 15 cells (five splits, three dimensions) of every condition of the 165-cell simulator gate (0 = no cell responds, 1 = every cell); the values of Table 6. Circuit and kernel hashes are the provenance sensors; the semantic-kernel column is the anchored comparison against $\Phi(C_0)$. Coverage is conditional on the evidence exposed to each sensor; output invariance does not imply workflow integrity, and the shot rows are binomial emulation, not QPU runs.

motivates the calibrated treatment of (iii), not a universal law. In this ideal-statevector gate the honest estimate equals the semantic kernel and g is the identity, so the single anchored comparison against $\Phi(C_0)$ acts as the semantic probe on class (a) and as the observed-kernel reference on class (c).

All nine frozen acceptance checks pass: 165 cells are present; clean evidence is invariant; benign transpilation and the common-unitary rewrite alter provenance but preserve semantic kernels and outputs; every data-dependent circuit mutation changes the semantic kernel; symmetry and diagonal sensors catch their respective malformed edits; the PSD-preserving mixture passes algebra but is exposed by trusted provenance/semantics; and both shot estimators produce non-zero repeat discrepancy in every cell. The shot conditions sample exact fidelities: they exercise the contract under controlled stochastic estimation but are neither sampler-backend nor QPU runs. PSD repair is containment, not proof of integrity.

TABLE 7
Composition of the frozen four-contract envelopes with the policy layer under $\mathcal{I}^{\star}_{XFY}$ (lattice maximum). Approved rewrites and approved stochastic estimation are not exact violations; the contract's own `hold` for the approved emulator is preserved.

Scenario	Contract	Exact viol.	Policy	Composed
Clean	`allow`	no	`allow`	`allow`
Approved transpilation	`allow`	no	`allow`	`allow`
Parameterized circuit mutation	`block`	yes	`block`	`block`
PSD-preserving kernel substitution	`block`	yes	`block`	`block`
Prior-preserving label corruption	`block`	yes	`block`	`block`
Approved 256-shot emulator	`hold`	no	`allow`	`hold`

9 Executable Contract and Policy Composition

The prototype evaluates four ordered contracts: input/preprocessing (input array and preprocessing declaration hashes; mismatch blocks); circuit/kernel (circuit and kernel provenance, semantic probes, symmetry, diagonal, eigenvalue and PSD-repair evidence; an equivalent rewrite passes only if explicitly approved); execution/result (declared execution metadata, repeated-estimation discrepancy and prediction provenance; approved stochastic execution is held); evaluation/report (item-level labels bound to predictions and balanced accuracy recomputed; prior-preserving corruption blocks even when the marginal is identical). Contracts form an ordered SHA-256 chain: later modification is detectable, but the chain is not signed and does not authenticate a provider. The contracts fail closed on their invariants: all passes yield `allow`; at least one review and no failure yields `hold`; any failure yields `block`. Six prespecified scenarios and all six chains pass eight strict acceptance checks in the frozen artifact. The policy layer composes with these contracts by the maximum in the lattice; the composition over the six frozen envelopes is Table 7, where an exact violation is derived from the recorded evidence with the approval flags respected. The policy layer is a pure function evaluated offline on frozen outputs; nothing in this article is a deployed runtime service.

10 Preregistered Reinforcement Gates (Artifact 1.1.0)

Three sensitivity gates were added after the 1.0.0 evidence was frozen. Their protocol, endpoints and one implementation clarification (amendment A1) were recorded in `manuscript/paper15_v11_reinforcement_` `prereg.md` before execution; results are reported as executed. The gates reuse the exact-statevector engine, the frozen split and model seeds and the frozen 18-condition suite.

10.1 Gate N: Null Calibration with Disjoint Clean Pools

For each of the 240 expansion runs the clean pool is the set of evaluation-side rows not selected into the frozen evaluation batch ($900-n$ in ID protocols, $3{,}000-n$ in OOD protocols), transformed with the same training-fitted projection and scaler as the frozen run. A permutation seeded by the split seed divides the pool into a calibration half and an evaluation half; twenty clean batches of n rows are drawn from each half by simple random sampling without replacement; the suite also includes an identity control and two near-null controls (Gaussian noise $\sigma = 0.001$, scaling $\alpha = 0.001$) applied in place. Ten distributional sensors are calibrated per (environment, dimension, model) cell at the 191st of 200 pooled calibration values ($\alpha = 0.05$ per sensor; fire if strictly greater); thresholds are frozen before any attacked row is examined; the empirical false-alarm rate is the firing fraction on the 200 disjoint evaluation draws. Two auditors are distinguished (amendment A1): the batch-level auditor uses only calibrated distributional sensors, and the item-aligned auditor compares the same items against trusted references with an exact-zero null. In E1 ($n = 256$) each pool half holds 322 rows, so the draws overlap almost entirely and E1 rates are indicative only.

Results. All 240 runs completed (14/14 acceptance checks). Table 8 reports the pooled per-sensor false-alarm rates (0.029–0.069 against the nominal 0.05) and the uncorrected regime unions; Table 9 the environment dependence; Table 10 the calibrated detection rates and the near-null control responses; Table 11 the label path (calibrated $\mathcal{I}_X$, $\mathcal{I}_{XF}$ and prior-preserving $\mathcal{I}_{Y_m}$ fire on none of the 3,600 label observations; the batch-level $\mathcal{I}_{XFY}$ union fires on 43; the item-aligned auditor detects all 3,600); and Table 12 the cluster structure that makes the KS sensor react to in-place perturbations of order 10^{-3} standard deviations while its between-batch null stays narrow, the fingerprint that Gate A attacks. The intervals in Table 8 are descriptive.

TABLE 8
Gate N: empirical false-alarm rate of the null-calibrated sensors and batch-level regimes on 12,000 disjoint clean evaluation draws (all eight environments pooled; nominal $\alpha = 0.05$ per sensor). Intervals are descriptive Clopper–Pearson 95% intervals: draws within a cell share a pool half and are not independent. Regime rows are unions without correction.

Sensor / regime	Fires	Rate	95% CI
Feature JSD	600	0.050	[0.046, 0.054]
Feature MMD	822	0.069	[0.064, 0.073]
KS rejection rate	342	0.029	[0.026, 0.032]
Score JSD	668	0.056	[0.052, 0.060]
Prediction-rate shift	520	0.043	[0.040, 0.047]
Prediction JSD	586	0.049	[0.045, 0.053]
Label prior shift	570	0.048	[0.044, 0.051]
Label JSD	570	0.048	[0.044, 0.051]
Confusion profile L_1	517	0.043	[0.040, 0.047]
Confusion profile JSD	638	0.053	[0.049, 0.057]
$\mathcal{I}_X$ (any sensor)	1500	0.125	[0.119, 0.131]
$\mathcal{I}_{XF}$ (any sensor)	2438	0.203	[0.196, 0.210]
$\mathcal{I}_{Y_m}$ (any sensor)	570	0.048	[0.044, 0.051]
$\mathcal{I}_{XFY}$ (any sensor)	3113	0.259	[0.252, 0.267]

10.2 Gates P and T: Preprocessing Ablation and Tuned Baselines

The CICIDS ID design (128/128, paper-core, ZZ and SVC, five splits, two model seeds, three dimensions) is re-executed under three scaler configurations (P0 reference: standard for SVC, min–max to 2π for QSVC; PA standard for both; PB min–max for both) and with both learners tuned on training rows only (SVC over $C \in \{0.1, 1, 10, 100\} \times \gamma \in \{\text{scale}, 0.01, 0.1, 1\}$, QSVC over $C \in \{0.1, 1, 10, 100\}$ on the precomputed training kernel, stratified five-fold cross-validation, balanced-accuracy scoring, ties toward the smallest C) in CICIDS ID and in the OOD environment with the highest mean clean SVC balanced accuracy in the frozen tables (UNSW temporal OOD, E8, 0.753). Table 13 reports the paired differences and Table 14 the clean balanced accuracies and selected constants. The direction of the secondary profile is stable (five of five positive split clusters in every configuration, dimension and environment); its magnitude is conditional: standardizing the quantum branch reduces the difference to roughly a third, matched min–max scaling leaves it unchanged, tuning widens it in CICIDS ID as ZZ gains clean headroom and leaves UNSW temporal OOD unchanged. Exact label-path invariance holds in every configuration.

11 Preregistered Policy Gates (Artifacts 1.2.0 and 1.3.0)

Two gates were preregistered on 2026-09-07 in `manuscript/paper15_v12_policy_prereg.md` before any analysis and computed from the frozen 1.1.1 outputs without re-executing any kernel or draw. Amendment A1 records the finding that the 1.1.x column `impact_bal_acc` is the positive part of the signed conclusion change, so that the material label-path count under the prespecified $|\Delta_R| > 0$ definition is 2,617 (2,184 lowered, 433 raised) rather than the 2,184 lowered rows alone. Amendment A2 (`manuscript/paper15_v13_prereg.md`, artifact 1.3.0) records that the family rule executed in 1.2.0 was asymmetric and that its stated level was false (Proposition 5(c)), adopts the conformal rule of Proposition 5(b), fixes the expectations before regeneration, and discloses that a diagnostic computation of the conformal rule had been seen during the review that found the defect (pooled rates 0.056 / 0.058 / 0.048 / 0.053, which the regeneration reproduces exactly). Both gates were regenerated from the same frozen outputs; the 1.2.0 numbers are retained as comparison columns and are used for no claim.

11.1 Gate F: Conformal Family Calibration

For every cell and regime the conformal p-value of a batch is computed as in Proposition 5(b) from the 200 calibration draws of the cell, and the regime fires when $p \leq 0.05$. The frozen per-sensor thresholds are reproduced from the calibration draws as a consistency check (F0). Table 15 gives the decision-level false-alarm rates of the union rule, the superseded 1.2.0 rule and the conformal rule in the

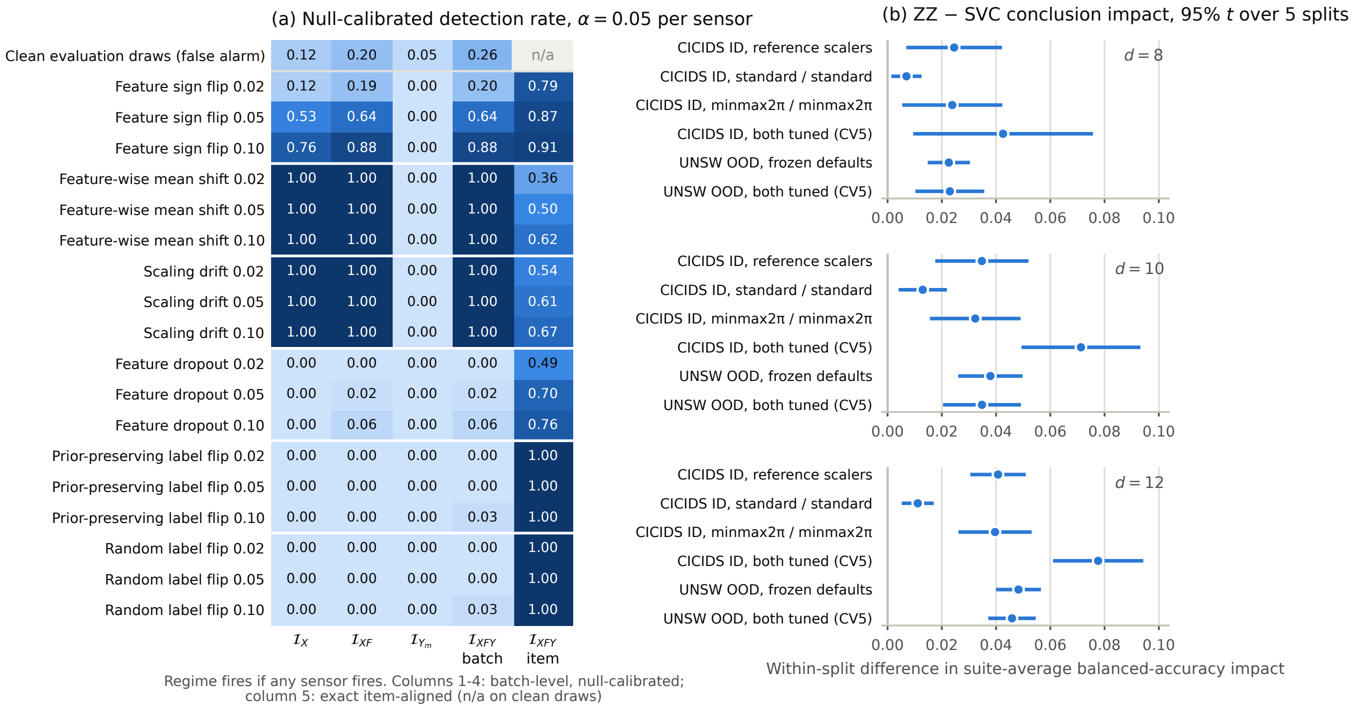


Fig. 2. Preregistered gates of artifact 1.1.0. (a) Null-calibrated detection rate of each information regime on the frozen interventions with the empirical false-alarm rate of disjoint clean evaluation draws in the first row; the first four columns are batch-level unions of per-sensor calibrated rules, the last column is exact item-aligned detection (undefined on clean draws). (b) Secondary within-split ZZ-minus-SVC conclusion-impact difference with 95% t intervals over five split clusters under three scaler configurations and under cross-validated tuning of both learners.

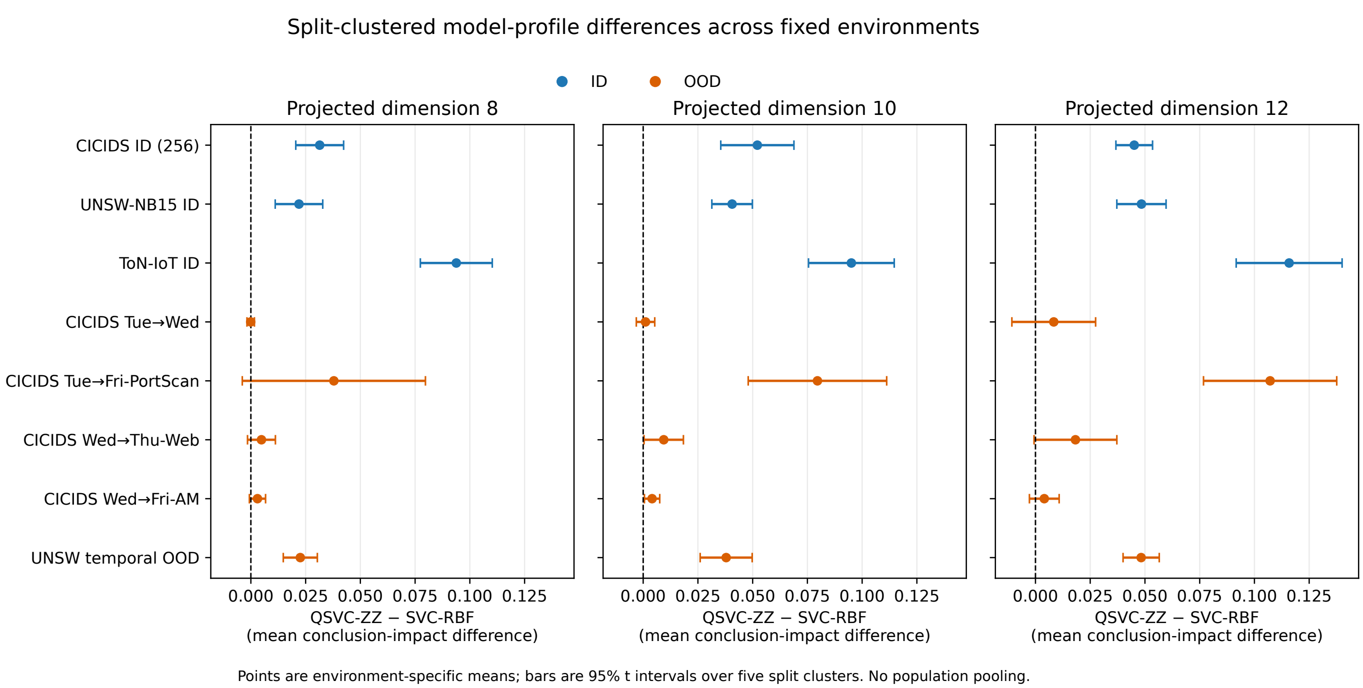


Fig. 3. Environment-specific ZZ-minus-SVC mean conclusion-impact differences and 95% t intervals over five split clusters. ID and OOD settings are fixed cases; no population pooling is performed.

TABLE 9
Gate N: per-environment false-alarm rates on 1,200 evaluation draws (three dimensions, two or four models, 200 draws per cell). Sensor range is the minimum and maximum per-sensor rate in the environment; regime columns are batch-level unions. E1 draws 256 rows from 322-row pool halves and is indicative only.

Environment	n	Pool half	Sensor range	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$
E1 CICIDS ID 256	256	322	0.023–0.130	0.203	0.245	0.045	0.331
E2 UNSW-NB15 ID	128	386	0.008–0.070	0.063	0.135	0.070	0.210
E3 ToN-IoT ID	128	386	0.025–0.172	0.215	0.280	0.070	0.358
E4 Tue→Wed	128	1436	0.025–0.082	0.150	0.230	0.025	0.254
E5 Tue→Fri-PortScan	128	1436	0.018–0.051	0.066	0.160	0.025	0.198
E6 Wed→Thu-Web	128	1436	0.013–0.057	0.070	0.163	0.025	0.181
E7 Wed→Fri-AM	128	1436	0.025–0.066	0.133	0.212	0.025	0.258
E8 UNSW temporal OOD	128	1436	0.032–0.092	0.072	0.192	0.050	0.237

TABLE 10
Gate N: detection rate of each regime on the frozen `paper_core` observations (all environments, dimensions and models pooled; 600 observations per intervention row) at the null-calibrated thresholds, and firing rate on the in-place near-null controls. Batch-level regimes use calibrated distributional sensors; item-aligned regimes use exact zero thresholds on item-wise deltas.

Intervention	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$	$\mathcal{I}_{XF}$ item	$\mathcal{I}_{XFY}$ item
Feature sign flip 0.02	0.118	0.192	0.000	0.200	0.793	0.793
Feature sign flip 0.05	0.532	0.640	0.000	0.643	0.867	0.867
Feature sign flip 0.10	0.757	0.877	0.000	0.878	0.907	0.907
Mean shift 0.02	1.000	1.000	0.000	1.000	0.357	0.357
Mean shift 0.05	1.000	1.000	0.000	1.000	0.500	0.500
Mean shift 0.10	1.000	1.000	0.000	1.000	0.618	0.618
Scaling drift 0.02	1.000	1.000	0.000	1.000	0.537	0.537
Scaling drift 0.05	1.000	1.000	0.000	1.000	0.608	0.608
Scaling drift 0.10	1.000	1.000	0.000	1.000	0.668	0.668
Feature dropout 0.02	0.000	0.000	0.000	0.000	0.488	0.488
Feature dropout 0.05	0.000	0.022	0.000	0.022	0.705	0.705
Feature dropout 0.10	0.000	0.062	0.000	0.062	0.763	0.763
Prior-preserving label flip 0.02	0.000	0.000	0.000	0.000	0.000	1.000
Prior-preserving label flip 0.05	0.000	0.000	0.000	0.002	0.000	1.000
Prior-preserving label flip 0.10	0.000	0.000	0.000	0.033	0.000	1.000
Random label flip 0.02	0.000	0.000	0.000	0.000	0.000	1.000
Random label flip 0.05	0.000	0.000	0.000	0.003	0.000	1.000
Random label flip 0.10	0.000	0.000	0.000	0.033	0.000	1.000
Near-null Gaussian $\sigma = 0.001$	0.713	0.713	0.000	0.713	0.042	0.042
Near-null scaling $\alpha = 0.001$	0.879	0.882	0.000	0.882	0.099	0.099

TABLE 11
Gate N: label-path interventions in the frozen expansion; number of observations on which each regime fires after calibration (batch-level) or by exact item-aligned comparison.

Subset	n	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$	item-aligned
All label interventions	3,600	0	0	0	43	3,600
Prior-preserving	1,800	0	0	0	21	1,800
Positive conclusion impact	2,184	0	0	0	40	2,184

TABLE 12
Cluster structure of the frozen evaluation batches (mean over the 30 cells of each environment, standardized projected features): fraction of near-duplicate rows, and the largest fraction of rows within a window of 0.01 or 0.001 standard deviations in any feature (max) or on average over features (mean).

Environment	Dup. rows	Max 0.01	Mean 0.01	Max 0.001
E1 CICIDS ID 256	0.074	0.825	0.562	0.648
E2 UNSW-NB15 ID	0.114	0.735	0.458	0.514
E3 ToN-IoT ID	0.257	0.870	0.466	0.571
E4 Tue→Wed	0.054	0.766	0.489	0.480
E5 Tue→Fri-PortScan	0.129	0.872	0.747	0.822
E6 Wed→Thu-Web	0.004	0.827	0.433	0.696
E7 Wed→Fri-AM	0.026	0.769	0.670	0.663
E8 UNSW temporal OOD	0.100	0.694	0.437	0.477

primary aggregate over the eight environments, in the sensitivity aggregate without E1 and in E1 alone; Table 16 the decomposition of the 1.2.0 excess into rule bias and design effect together with both rules under exchangeable re-splits; Table 17 the split construction S1; Tables 18 and 19 the per-environment cluster means with five-cluster t intervals; and Table 20 the detection rates on the frozen interventions and the near-null controls. The conformal $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ rules fire on none of the 3,600 label observations and the $\mathcal{I}_{Y_m}$ rule on none of the 1,800 prior-preserving ones (F3); the batch-level $\mathcal{I}_{XFY}$ conformal rule fires on 11 of the 2,617 material label rows (1.2.0 rule 16, union 43). Under the re-splits the conformal rule stays at or below its level in every regime (F4) and the 1.2.0 rule does not.

11.2 Gate D: Offline End-to-End Decisions

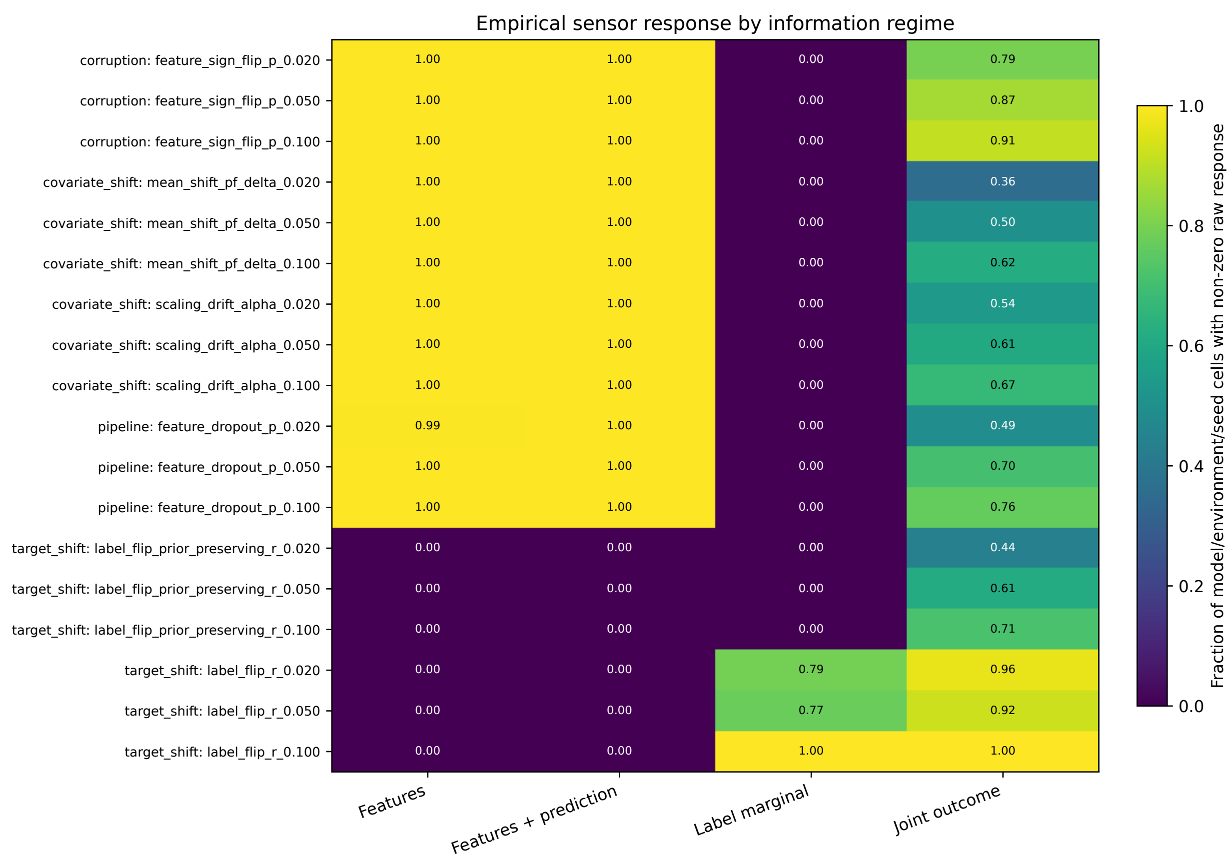
Four policies (P0 serve-always, baseline; P1 the uncalibrated union of per-sensor rules, risk-tolerant; P2 the conformal family rule, calibrated and risk-tolerant; P3 the coverage-complete abstaining variant, which holds whenever a mandatory protected boundary has neither exact nor declared

TABLE 13
Gates P and T: within-split ZZ-minus-SVC difference in suite-average balanced-accuracy conclusion impact, mean with 95% t interval over five split clusters and number of positive clusters. CICIDS ID reference is the re-executed P0 configuration; UNSW OOD reference is the frozen 1.0.0 environment E8.

Condition	$d = 8$	$d = 10$	$d = 12$
P0 reference (standard / minmax2π)	0.0245 [0.0069, 0.0422] (5/5)	0.0348 [0.0176, 0.0520] (5/5)	0.0407 [0.0305, 0.0509] (5/5)
PA standard / standard	0.0070 [0.0014, 0.0125] (5/5)	0.0130 [0.0041, 0.0219] (5/5)	0.0111 [0.0052, 0.0170] (5/5)
PB minmax2π / minmax2π	0.0238 [0.0054, 0.0423] (5/5)	0.0323 [0.0156, 0.0491] (5/5)	0.0396 [0.0261, 0.0531] (5/5)
CICIDS ID, both tuned (CV5)	0.0426 [0.0094, 0.0757] (5/5)	0.0713 [0.0494, 0.0932] (5/5)	0.0776 [0.0610, 0.0943] (5/5)
UNSW OOD, frozen defaults	0.0226 [0.0148, 0.0304] (5/5)	0.0379 [0.0260, 0.0498] (5/5)	0.0483 [0.0400, 0.0565] (5/5)
UNSW OOD, both tuned (CV5)	0.0229 [0.0103, 0.0356] (5/5)	0.0348 [0.0205, 0.0492] (5/5)	0.0459 [0.0371, 0.0546] (5/5)

TABLE 14
Gates P and T: clean balanced accuracy (SVC / ZZ, mean over five split clusters) per configuration and dimension, and the distribution of cross-validated regularisation constants over the 30 cells of each tuned environment (SVC also selects γ; see the artifact table).

Condition	$d = 8$	$d = 10$	$d = 12$
P0 reference (standard / minmax2π)	0.753 / 0.734	0.747 / 0.728	0.778 / 0.747
PA standard / standard	0.753 / 0.727	0.747 / 0.720	0.778 / 0.727
PB minmax2π / minmax2π	0.751 / 0.734	0.750 / 0.728	0.756 / 0.747
CICIDS ID, both tuned (CV5)	0.741 / 0.750	0.746 / 0.781	0.780 / 0.802
UNSW OOD, frozen defaults	0.748 / 0.760	0.754 / 0.766	0.755 / 0.769
UNSW OOD, both tuned (CV5)	0.739 / 0.759	0.723 / 0.757	0.736 / 0.770
Tuned environment	**QSVC C selections**		**SVC C selections**
CICIDS ID	$C = 0.1$: 3, $C = 1$: 4, $C = 10$: 6, $C = 100$: 17		$C = 0.1$: 1, $C = 1$: 6, $C = 10$: 15, $C = 100$: 8
UNSW OOD	$C = 1$: 17, $C = 10$: 13		$C = 0.1$: 1, $C = 1$: 2, $C = 10$: 15, $C = 100$: 12



A zero is a raw invariance result, not an estimated detector power; practical alarms require calibration.

Fig. 4. Raw sensor response by information regime in the frozen expansion. Fractions are over model/environment/seed cells; zero denotes raw invariance, not estimated detector power.

statistical coverage and which guarantees no minimum detection power on the boundaries that are covered) are evaluated over five regimes on 12,000 clean evaluation draws (the 1,200 exact-zero rows for $\mathcal{I}^{\star}_{XFY}$), 1,200 near-

TABLE 15
Gate F: decision-level false-alarm rate on the disjoint clean evaluation draws (12,000 in the primary aggregate; nominal $\alpha = 0.05$ per decision; exact level of the conformal rule $10/201 = 0.0498$) of the union of per-sensor rules (1.1.0), of the superseded asymmetric family rule (1.2.0) and of the adopted conformal family rule (1.3.0). Pooled rates are descriptive because draws within a cell overlap; the last column is the range of the per-environment cluster means of the conformal rule (five split clusters each). E1 is never excluded from the primary aggregate.

Regime	Union	1.2.0 rule	Conformal	Env. cluster means (conformal)
All eight environments (primary)				
$\mathcal{I}_X$	0.125	0.061	0.056	0.023–0.125
$\mathcal{I}_{XF}$	0.203	0.073	0.058	0.023–0.115
$\mathcal{I}_{Y_m}$	0.048	0.048	0.048	0.025–0.070
$\mathcal{I}_{XFY}$	0.259	0.079	0.053	0.034–0.123
Seven environments, E1 excluded (sensitivity)				
$\mathcal{I}_X$	0.116	0.053	0.048	0.023–0.092
$\mathcal{I}_{XF}$	0.199	0.067	0.052	0.023–0.081
$\mathcal{I}_{Y_m}$	0.048	0.048	0.048	0.025–0.070
$\mathcal{I}_{XFY}$	0.251	0.071	0.045	0.034–0.065
E1 only (256-row draws from 322-row halves)				
$\mathcal{I}_X$	0.203	0.133	0.125	0.125–0.125
$\mathcal{I}_{XF}$	0.245	0.125	0.115	0.115–0.115
$\mathcal{I}_{Y_m}$	0.045	0.045	0.045	0.045–0.045
$\mathcal{I}_{XFY}$	0.331	0.146	0.123	0.123–0.123

TABLE 16
Gate F: decomposition of the pooled excess of the 1.2.0 rule over the nominal 0.05 into rule bias (1.2.0 rate minus conformal rate on the same draws) and design effect (conformal rate minus nominal), and both rules under 30 exchangeable random re-splits of the 400 pooled draws of every cell (seed 20260907), where the roles are assigned by a uniform random permutation and any excess of the 1.2.0 rule is its own bias; the conformal rule must stay at or below $10/201 = 0.0498$.

Regime	1.2.0	Conformal	Rule bias	Design	Re-split 1.2.0	Re-split conf.
$\mathcal{I}_X$	0.061	0.056	+0.005	+0.006	0.052	0.044
$\mathcal{I}_{XF}$	0.073	0.058	+0.015	+0.008	0.058	0.042
$\mathcal{I}_{Y_m}$	0.048	0.048	+0.000	-0.003	0.040	0.040
$\mathcal{I}_{XFY}$	0.079	0.053	+0.025	+0.003	0.059	0.036

TABLE 17
Sensitivity S1: split construction (reference draws 01–10 of each run define the rank transform, calibration draws 11–20 give the null family scores; p-value resolution $1/101$) on the same 12,000 evaluation draws. Valid under the same premise but using half of the draws for each role; the adopted rule is the full conformal one.

Regime	Draws	Fires	Pooled rate
$\mathcal{I}_X$	12,000	592	0.049
$\mathcal{I}_{XF}$	12,000	470	0.039
$\mathcal{I}_{Y_m}$	12,000	804	0.067
$\mathcal{I}_{XFY}$	12,000	403	0.034

null synthetic controls and the 10,800 frozen intervened observations, of which 7,008 are material at $|\Delta_R| > 0$ (1,870 at $\tau = 0.05$). These form 24,000 policy rows derived from the frozen intervention grid; inferential independence resides at the prespecified environment/split clusters rather than at row level. The 12,000 batch clean-decision draws and 1,200 trusted exact-zero invariant rows are different estimands and are never pooled. `block` is reserved for exact violations against a trusted reference; the interruption of the near-null controls is therefore decomposed into statistical holds and exact-reference blocks. In the trusted regime P2 produces 85 gross exact blocks, of which 46 overlap the batch $\mathcal{I}_{XFY}$/P2 interruptions; its net increment is 39/1,200 (3.25 percentage points), and the trusted total is 629 of 1,200. Table 21 gives the complete decision counts including the hold and block counts on the near-null controls and the non-material interruption rate, Table 22 the calibrated policy under the superseded and the adopted rule, Table 23 the materiality sensitivity and Table 24 the materially altered audit results served by mechanism under P2. The five consistency checks D1–D5 (serve-always allows every material observation; structural blindness is reproduced by the decisions; the trusted regime blocks every material label row and every exact violation; decisions are total and deterministic; the composition reproduces the frozen contract actions) all pass.

TABLE 24
Gate D: materially altered audit results served / material observations by intervention mechanism under the conformal calibrated policy P2 (structural-sensitivity endpoint $|\Delta_R| > 0$).

Regime	Evaluation-label	Sign flip	Mean shift / scaling	Feature dropout
$\mathcal{I}_X$	2,617/2,617	800/1,464	0/1,848	1,079/1,079
$\mathcal{I}_{XF}$	2,617/2,617	685/1,464	6/1,848	1,057/1,079
$\mathcal{I}_{Y_m}$	2,617/2,617	1,464/1,464	1,848/1,848	1,079/1,079
$\mathcal{I}_{XFY}$	2,606/2,617	717/1,464	10/1,848	1,057/1,079
$\mathcal{I}^{\star}_{XFY}$	0/2,617	0/1,464	0/1,848	0/1,079

12 Preregistered Adversarial Gate A (Artifact 1.3.0)

Gate A was preregistered in `manuscript/paper15_v13_prereg.md` before execution: threat knowledge, perturbation algorithm, cluster rule (an entry is frozen when at least $\lceil 0.05n \rceil$ rows, never fewer than two, lie within 0.01 batch standard deviations of it; constant features frozen), budget (the unclustered entries, data-determined), strength grid (0.02, 0.05, 0.10 matched to the frozen suite; 0.25, 0.50), matched controls (the executed mechanisms with their frozen tags and seeds), environments, models, seeds, endpoints, acceptance checks, expectations and stopping rule. The 240 jobs completed; the clean rows and matched controls reproduce the frozen expansion exactly (AC1); labels are untouched (AC2); the frozen-entry fraction is recorded on every adaptive row (AC3); every material adaptive row carries a prediction change (AC4). Scoring uses the calibration draws, thresholds, rules and policy layer of Gates F and D. Table 25 gives the compact overview at matched strengths, Table 27 the detection rates under the three rules and the exact item-aligned detection, Table 28 the prediction-change and material fractions with the attacker's budget, Table 26 the complete per-strength materiality, detection and P2 served profile, Table 29 the materially altered audit results served under P1–P3, Table 30 the per-environment detection in $\mathcal{I}_{XF}$ and Table 31 the classical and quantum branches separately. The expectation recorded before execution (detection below 0.30 at matched strengths) held for the mean shift in $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ at 0.02 and 0.05 and for scaling drift in $\mathcal{I}_X$ at 0.02 and 0.05, and did not hold at 0.10 (0.34–0.66); it is reported as executed. No parameter was changed after the results were seen.

TABLE 18
Gate F: per-environment false-alarm rate of the conformal family rule (cluster mean over the five split clusters with the two-sided 95% t interval, four degrees of freedom). n is the batch size; "disjoint" is the number of mutually disjoint batches that one pool half can supply, an upper bound on the effective number of independent draws among the 20 per run (E1: one). The union and 1.2.0 cluster means are in Table 19.

Environment	n	disj.	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$
E1 CICIDS ID 256	256	1	0.125 [-0.05,0.30]	0.115 [-0.03,0.26]	0.045 [-0.05,0.14]	0.123 [0.00,0.24]
E2 UNSW-NB15 ID	128	3	0.023 [-0.02,0.06]	0.042 [0.01,0.07]	0.070 [0.01,0.13]	0.035 [0.01,0.06]
E3 ToN-IoT ID	128	3	0.065 [-0.02,0.15]	0.075 [0.01,0.15]	0.070 [0.01,0.13]	0.062 [0.01,0.12]
E4 CICIDS Tue→Wed	128	11	0.092 [-0.01,0.19]	0.081 [0.01,0.15]	0.025 [-0.01,0.06]	0.065 [0.02,0.11]
E5 CICIDS Tue→Fri-PortScan	128	11	0.032 [-0.01,0.08]	0.023 [-0.02,0.06]	0.025 [-0.01,0.06]	0.037 [-0.00,0.08]
E6 CICIDS Wed→Thu-Web	128	11	0.044 [0.00,0.09]	0.036 [-0.01,0.08]	0.025 [-0.01,0.06]	0.037 [0.00,0.07]
E7 CICIDS Wed→Fri-AM	128	11	0.049 [0.01,0.09]	0.041 [0.02,0.06]	0.025 [-0.01,0.06]	0.034 [0.02,0.05]
E8 UNSW temporal OOD	128	11	0.043 [-0.03,0.12]	0.049 [0.02,0.08]	0.050 [-0.01,0.11]	0.041 [0.03,0.05]

TABLE 19
Gate F: per-environment cluster means of the union rule / the superseded 1.2.0 rule, for comparison with Table 18 (environment codes as there).

Env.	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$
E1	0.203 / 0.133	0.245 / 0.125	0.045 / 0.045	0.331 / 0.146
E2	0.063 / 0.025	0.135 / 0.051	0.070 / 0.070	0.210 / 0.054
E3	0.215 / 0.071	0.280 / 0.088	0.070 / 0.070	0.358 / 0.102
E4	0.150 / 0.098	0.230 / 0.095	0.025 / 0.025	0.254 / 0.102
E5	0.066 / 0.039	0.160 / 0.045	0.025 / 0.025	0.197 / 0.046
E6	0.070 / 0.044	0.163 / 0.054	0.025 / 0.025	0.181 / 0.054
E7	0.133 / 0.060	0.212 / 0.063	0.025 / 0.025	0.257 / 0.061
E8	0.072 / 0.043	0.192 / 0.065	0.050 / 0.050	0.237 / 0.065

TABLE 29
Gate A: materially altered audit results served under P1 union, P2 conformal and P3 coverage-complete abstaining per regime, summed over mechanisms and strengths (structural-sensitivity endpoint $|\Delta_R| > 0$). Under $\mathcal{I}^{\star}_{XFY}$ every material row carries a prediction change and is blocked.

Regime	Material	P1 union	P2 conformal	P3 abst.
Executed mechanisms (1,800 rows)				
$\mathcal{I}_X$	1,848	0	0	0
$\mathcal{I}_{XF}$	1,848	0	6	0
$\mathcal{I}_{XFY}$	1,848	0	10	10
$\mathcal{I}^{\star}_{XFY}$	1,848	0	0	0
Cluster-preserving F5 (6,000 rows)				
$\mathcal{I}_X$	3,418	1,207	1,328	0
$\mathcal{I}_{XF}$	3,418	719	959	0
$\mathcal{I}_{XFY}$	3,418	716	1,006	1,006
$\mathcal{I}^{\star}_{XFY}$	3,418	0	0	0

TABLE 31
Gate A: detection of the adaptive variants under the conformal rule by branch (classical SVC on standardized features; fidelity kernels on features scaled to $[0, 2\pi]$). The cluster rule is the same in both branches; the branch scaler changes how much the unclustered entries move relative to the null.

		Classical		Quantum	
Mechanism	Strength	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_X$	$\mathcal{I}_{XF}$
Mean shift	0.02	0.00	0.00	0.01	0.10
Mean shift	0.05	0.04	0.04	0.10	0.37
Mean shift	0.10	0.30	0.34	0.37	0.58
Mean shift	0.25	0.72	0.72	0.84	0.85
Mean shift	0.50	0.85	0.86	0.91	0.93
Scaling drift	0.02	0.00	0.00	0.08	0.38
Scaling drift	0.05	0.14	0.14	0.49	0.69
Scaling drift	0.10	0.33	0.39	0.72	0.84
Scaling drift	0.25	0.47	0.57	0.94	0.94
Scaling drift	0.50	0.72	0.75	0.97	0.97

13 Secondary Model-Impact Profile

The ZZ fidelity kernel has the largest mean conclusion impact in the frozen Gate-1 ID suite. Relative to SVC-RBF, mean impact ratios are 2.524, 3.094 and 3.323 at dimensions 8, 10 and 12; paired within-split differences are 0.02427 (95% CI 0.01200 to 0.03654), 0.03289 (0.01998 to 0.04580) and 0.03855 (0.02773 to 0.04937), with all five split clusters positive at every dimension (ZZ/SVC impacts 0.04019/0.01592, 0.04860/0.01571 and 0.05514/0.01659). Across the eight expansion environments the mean ZZ-minus-SVC impact is positive in 7/8 environments at dimension 8 and 8/8 at dimensions 10 and 12; environment-specific intervals exclude zero in 4/8, 7/8 and 5/8; means range from -0.00009 to 0.09382, 0.00107 to 0.09508 and 0.00403 to 0.11585 (Fig. 3). Clean balanced accuracy and impact correlate positively across expansion cells ($r = 0.714$, exploratory). The preregistered ablations (Fig. 2(b)) show that the direction is stable but the magnitude is conditional: standardizing the quantum branch shrinks the difference to about a third, matched min–max scaling leaves it unchanged, and cross-validated tuning of both learners widens it in CICIDS ID and leaves UNSW temporal OOD unchanged. No pooled population effect is claimed, the profile is neither a quantum-advantage nor a vulnerability ordering, and the conditional-auditability result does not depend on it. Figure 4 shows the raw (uncalibrated) sensor response fractions by regime that motivated the calibration gates.

14 Reproduction and Evidence Manifest

The recommended read-only verification is

```
python -m src.experiments.verify_publication_artifact
       --root publication/artifact
```

It verifies the artifact-wide SHA-256 manifest, the eight embedded evidence manifests and their outputs, row counts, the primary-claim counts including the signed label-path counts, the calibrated label-path consistency checks of the reinforcement gate, the policy-level claims of the regenerated gates (including the exact level of the conformal rule under the exchangeable re-splits), and the replay and trust checks of the adversarial gate. The complete test command is

```
python -m pytest tests -q -p no:cacheprovider
```

and includes the exhaustive counting-bound tests of Proposition 5(b), the five-vector counterexample to the 1.2.0 rule, the brute-force checks of Lemma 1, Propositions 1–4, 6 and 7 and Corollaries 1–3 on enumerated batches, the intervention-recomputation semantics of the workflow state, the finite-shot counterexample of Proposition 7(iii), and the

TABLE 20
Gate F: detection rate on the frozen `paper_core` observations (600 per intervention row, eight environments pooled) and firing rate on the near-null in-place controls, as union rule / 1.2.0 rule / conformal rule. The conformal rule trades a small loss of detection for the decision-level false-alarm level of Table 15.

Intervention	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$
Feature sign flip 0.02	0.12 / 0.09 / 0.09	0.19 / 0.15 / 0.13	0.00 / 0.00 / 0.00	0.20 / 0.15 / 0.12
Feature sign flip 0.05	0.53 / 0.47 / 0.47	0.64 / 0.56 / 0.52	0.00 / 0.00 / 0.00	0.64 / 0.53 / 0.47
Feature sign flip 0.10	0.76 / 0.72 / 0.72	0.88 / 0.85 / 0.84	0.00 / 0.00 / 0.00	0.88 / 0.84 / 0.83
Mean shift 0.02	1.00 / 1.00 / 1.00	1.00 / 1.00 / 0.97	0.00 / 0.00 / 0.00	1.00 / 0.99 / 0.97
Mean shift 0.05	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	0.00 / 0.00 / 0.00	1.00 / 1.00 / 1.00
Mean shift 0.10	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	0.00 / 0.00 / 0.00	1.00 / 1.00 / 1.00
Scaling drift 0.02	1.00 / 0.99 / 0.99	1.00 / 0.99 / 0.96	0.00 / 0.00 / 0.00	1.00 / 0.98 / 0.95
Scaling drift 0.05	1.00 / 1.00 / 1.00	1.00 / 1.00 / 0.99	0.00 / 0.00 / 0.00	1.00 / 1.00 / 0.99
Scaling drift 0.10	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	0.00 / 0.00 / 0.00	1.00 / 1.00 / 1.00
Feature dropout 0.02	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00
Feature dropout 0.05	0.00 / 0.00 / 0.00	0.02 / 0.01 / 0.01	0.00 / 0.00 / 0.00	0.02 / 0.01 / 0.01
Feature dropout 0.10	0.00 / 0.00 / 0.00	0.06 / 0.04 / 0.03	0.00 / 0.00 / 0.00	0.06 / 0.03 / 0.03
Prior-preserving label flip 0.02	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00
Prior-preserving label flip 0.05	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00
Prior-preserving label flip 0.10	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.03 / 0.01 / 0.01
Random label flip 0.02	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00
Random label flip 0.05	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00
Random label flip 0.10	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.00 / 0.00 / 0.00	0.03 / 0.01 / 0.01
Near-null Gaussian $\sigma = 0.001$	0.65 / 0.49 / 0.45	0.65 / 0.44 / 0.41	0.00 / 0.00 / 0.00	0.65 / 0.43 / 0.41
Near-null scaling $\alpha = 0.001$	0.86 / 0.68 / 0.64	0.86 / 0.64 / 0.62	0.00 / 0.00 / 0.00	0.86 / 0.62 / 0.58

TABLE 21
Gate D: complete decision counts at the primary materiality threshold ($|\Delta_R| > 0$; 7,008 material and 3,792 non-material intervened observations; 1,200 near-null synthetic controls). P0 serve-always, P1 union, P2 conformal, P3 coverage-complete abstaining. Clean rows: 12,000 disjoint draws for batch-level regimes; the 1,200 exact-zero rows for $\mathcal{I}^{\star}_{XFY}$. B. hold/block: near-null controls held (statistical) or blocked (exact-reference violation). Integ.: non-material intervened observations held or blocked. Safe allow: clean, benign and non-material observations allowed. Interr.: non-material interruption rate, (false hold + false block + benign hold + benign block + integrity-only hold/block) over all non-material observations.

Regime	Pol.	Clean	F. hold	F. block	B. hold	B. block	Served	T. hold	T. block	Integ.	Imm. allow	Safe allow	Interr.
$\mathcal{I}_X$	P0	12,000	0	0	0	0	7,008	0	0	0	3,792	16,992	0.00
$\mathcal{I}_X$	P1	12,000	1,500	0	907	0	4,429	2,579	0	1,865	1,927	12,720	0.25
$\mathcal{I}_X$	P2	12,000	672	0	654	0	4,496	2,512	0	1,850	1,942	13,816	0.19
$\mathcal{I}_X$	P3	12,000	12,000	0	1,200	0	0	7,008	0	3,792	0	0	1.00
$\mathcal{I}_{XF}$	P0	12,000	0	0	0	0	7,008	0	0	0	3,792	16,992	0.00
$\mathcal{I}_{XF}$	P1	12,000	2,438	0	910	0	4,231	2,777	0	1,898	1,894	11,746	0.31
$\mathcal{I}_{XF}$	P2	12,000	695	0	622	0	4,365	2,643	0	1,830	1,962	13,845	0.19
$\mathcal{I}_{XF}$	P3	12,000	12,000	0	1,200	0	0	7,008	0	3,792	0	0	1.00
$\mathcal{I}_{Y_m}$	P0	12,000	0	0	0	0	7,008	0	0	0	3,792	16,992	0.00
$\mathcal{I}_{Y_m}$	P1	12,000	570	0	0	0	7,008	0	0	0	3,792	16,422	0.03
$\mathcal{I}_{Y_m}$	P2	12,000	570	0	0	0	7,008	0	0	0	3,792	16,422	0.03
$\mathcal{I}_{Y_m}$	P3	12,000	12,000	0	1,200	0	0	7,008	0	3,792	0	0	1.00
$\mathcal{I}_{XFY}$	P0	12,000	0	0	0	0	7,008	0	0	0	3,792	16,992	0.00
$\mathcal{I}_{XFY}$	P1	12,000	3,113	0	910	0	4,180	2,828	0	1,898	1,894	11,071	0.35
$\mathcal{I}_{XFY}$	P2	12,000	637	0	590	0	4,390	2,618	0	1,817	1,975	13,948	0.18
$\mathcal{I}_{XFY}$	P3	12,000	637	0	590	0	4,390	2,618	0	1,817	1,975	13,948	0.18
$\mathcal{I}^{\star}_{XFY}$	P0	1,200	0	0	0	0	7,008	0	0	0	3,792	6,192	0.00
$\mathcal{I}^{\star}_{XFY}$	P1	1,200	0	0	841	85	0	0	7,008	3,016	776	2,250	0.64
$\mathcal{I}^{\star}_{XFY}$	P2	1,200	0	0	544	85	0	0	7,008	2,944	848	2,619	0.58
$\mathcal{I}^{\star}_{XFY}$	P3	1,200	0	0	544	85	0	0	7,008	2,944	848	2,619	0.58

TABLE 22
Gate D: the calibrated risk-tolerant policy P2 under the superseded 1.2.0 family rule and under the adopted conformal rule, on identical observations ($|\Delta_R| > 0$). The 1.2.0 columns reproduce the published 1.2.0 numbers and are retained for comparison only.

	1.2.0 rule (superseded)				Conformal rule (adopted)			
Regime	Clean FPR	Benign	Served	Contain.	Clean FPR	Benign	Served	Contain.
$\mathcal{I}_X$	0.061	0.58	4,494	0.36	0.056	0.55	4,496	0.36
$\mathcal{I}_{XF}$	0.073	0.54	4,327	0.38	0.058	0.52	4,365	0.38
$\mathcal{I}_{Y_m}$	0.048	0.00	7,008	0.00	0.048	0.00	7,008	0.00
$\mathcal{I}_{XFY}$	0.079	0.53	4,322	0.38	0.053	0.49	4,390	0.37
$\mathcal{I}^{\star}_{XFY}$	0.000	0.55	0	1.00	0.000	0.52	0	1.00

TABLE 23
Gate D: materially altered audit results served (containment in parentheses) under three materiality thresholds on $|\Delta_R|$; material observations 7,008, 3,802 and 1,870. The $\tau \to 0^+$ column is a structural-sensitivity endpoint, whereas 0.02 and 0.05 are larger-effect sensitivity analyses. P1 union, P2 conformal.

Regime	Pol.	$\tau = 0^+$	$\tau = 0.02$	$\tau = 0.05$
$\mathcal{I}_X$	P1	4,429 (0.37)	2,128 (0.44)	652 (0.65)
$\mathcal{I}_X$	P2	4,496 (0.36)	2,174 (0.43)	680 (0.64)
$\mathcal{I}_{XF}$	P1	4,231 (0.40)	1,977 (0.48)	566 (0.70)
$\mathcal{I}_{XF}$	P2	4,365 (0.38)	2,074 (0.45)	629 (0.66)
$\mathcal{I}_{Y_m}$	P1	7,008 (0.00)	3,802 (0.00)	1,870 (0.00)
$\mathcal{I}_{Y_m}$	P2	7,008 (0.00)	3,802 (0.00)	1,870 (0.00)
$\mathcal{I}_{XFY}$	P1	4,180 (0.40)	1,926 (0.49)	521 (0.72)
$\mathcal{I}_{XFY}$	P2	4,390 (0.37)	2,088 (0.45)	629 (0.66)
$\mathcal{I}^{\star}_{XFY}$	P1	0 (1.00)	0 (1.00)	0 (1.00)
$\mathcal{I}^{\star}_{XFY}$	P2	0 (1.00)	0 (1.00)	0 (1.00)

manuscript consistency gates (abstract length, active headline numbers, conformal and policy wording, the interruption decomposition). The authoritative counts of tests, manifests, outputs, files, pages and hashes for the released version are generated into `publication/RELEASE_STATUS.md` by `src/experiments/make_release_status.py`. The frozen environment uses Python 3.10.16, NumPy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2, Qiskit 2.3.0 and Qiskit Machine Learning 0.9.0. Qiskit is pinned below 3 because migrating deprecated feature-map constructors would change circuit serialization and therefore requires a new evidence version. The artifact omits raw benchmark datasets; it documents their official sources, expected SHA-256 hashes, deterministic staging and a cross-platform hash verifier, and includes source code, locked dependencies, compact derived evidence, figures, figure-source tables, tests, manifests, preregistrations and expected outputs. The experimental evidence is frozen at artifact 1.3.0 (version DOI 10.5281/zenodo.22648573, tag `paper15-q1-v1.3.0`). Version 1.3.3 is archived at Zenodo under the concept DOI 10.5281/zenodo.22550852 with repository tag `paper15-q1-v1.3.3`; source code is licensed under Apache-2.0, derived evidence, figures and documentation under CC BY 4.0, and the manuscript files are author preprints excluded from both licenses.

15 Claims-Supported / Claims-Excluded Matrix

Supported by the artifact. Auditability is a property of (intervention class, view, trusted references); structural blind regions are monotone under refinement and closed exactly by added evidence that separates the class; feature/prediction sensors are exactly blind to label-only changes and label marginals to prior-preserving relabeling; every material label-path change alters the confusion matrix and is separable in the joint view, but batch-level Class-A auditors with a benchmark-protected, non-authenticated same-item-set statistical oracle detect at most 1.6% of them whereas a trusted aggregate reference of the same batch detects all of them and an item-aligned reference additionally exposes the 808 aggregate-preserving relabelings; the calibrated rules reproduce the structural blind regions (a consistency check of Proposition 4(i), not a finding); the union of per-sensor rules at $\alpha = 0.05$ yields decision false-alarm rates of 0.125–0.259 that the conformal family rule brings to 0.048–0.058 in the executed design, whose finite-sample level of $10/201$ holds under exchangeability, a premise the executed design violates by a measured, small margin concentrated in E1; the rule of artifact 1.2.0 had no valid guarantee and most of its published excess was rule bias; batch-level regimes with feature evidence serve 4,365–4,496 of 7,008 materially changed results under the calibrated risk-tolerant policy, the label-marginal regime serves all of them, and the trusted item-aligned regime serves none while interrupting 629 of 1,200 near-null synthetic controls (85 gross exact-reference blocks, but only 39 net additional interruptions versus batch $\mathcal{I}_{XFY}$/P2); an adaptive attacker who preserves the tight feature clusters cuts the batch-level detection of drift from 0.96–1.00 to 0.01–0.66 at matched strengths while keeping 83–91% of the conclusion changes, and its materially altered audit results are served by the calibrated policy in 28–39% of its material instances in aggregate, with the full profile reported by strength; the secondary ZZ-minus-SVC profile is direction-stable but scaler- and headroom-dependent; complementary quantum-workflow sensors close different simulator-gate blind regions as instances of one view lattice; and the local prototype executes policy-aware `allow/hold/block` composition offline on frozen outputs.

Explicitly excluded. The evidence does not estimate attack prevalence or population-level deployment incidence, establish universal superiority or vulnerability of quantum kernels, causally isolate feature-map geometry from preprocessing and clean headroom, establish that the exchangeability premise of the calibration holds in deployment, or claim a conditional guarantee given a calibration set. It evaluates one adaptive attacker model and no other. It provides no evidence about QPU security, calibrated device noise, scheduling, provider integrity or attestation, multi-tenancy, side channels, production readiness, deployed runtime services, non-repudiation, persistence, incident handling, recovery under non-stationarity, or operational Fleet Management.

TABLE 25

Gate A: executed drift mechanisms (matched controls, frozen strengths) against their adaptive cluster-preserving variants (class F5), 600 observations per row pooled over the eight environments. Detection uses the conformal family rule in $\mathcal{I}_X$ and $\mathcal{I}_{XF}$; material is the fraction with changed balanced accuracy; served is the number of undetected materially altered audit results served by P2 in $\mathcal{I}_{XF}$ over material rows. The attacker leaves clustered entries untouched and perturbs the remaining 32% on average.

		Executed mechanism (control)				Cluster-preserving (adaptive, F5)			
Mechanism	Strength	Det. $\mathcal{I}_X$	Det. $\mathcal{I}_{XF}$	Material	Served P2	Det. $\mathcal{I}_X$	Det. $\mathcal{I}_{XF}$	Material	Served P2
Mean shift	0.02	1.00	0.97	0.33	4/197	0.01	0.06	0.29	155/174
Mean shift	0.05	1.00	1.00	0.46	0/275	0.08	0.24	0.40	151/237
Mean shift	0.10	1.00	1.00	0.59	0/355	0.34	0.48	0.54	139/322
Mean shift	0.25	–	–	–	–	0.79	0.80	0.69	65/411
Mean shift	0.50	–	–	–	–	0.89	0.90	0.76	33/454
Scaling drift	0.02	0.99	0.96	0.48	2/290	0.05	0.23	0.40	131/241
Scaling drift	0.05	1.00	0.99	0.57	0/345	0.35	0.47	0.52	107/313
Scaling drift	0.10	1.00	1.00	0.64	0/386	0.56	0.66	0.58	76/350
Scaling drift	0.25	–	–	–	–	0.76	0.79	0.72	61/431
Scaling drift	0.50	–	–	–	–	0.88	0.88	0.81	41/485

TABLE 26

Adaptive F5 profile by nominal strength, derived without rerunning any experiment. Material is changed balanced accuracy at the structural-sensitivity endpoint. Detection is the conformal family firing rate over all 600 rows; served is the P2 fraction among material rows. The first three strengths have matched executed controls; 0.25 and 0.50 are the preregistered extended adaptive grid.

			Detection, all rows			Material served by P2		
Mechanism	Strength	Material	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{XFY}$	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{XFY}$
Mean shift	0.02	174/600	0.01	0.06	0.06	0.99	0.89	0.90
Mean shift	0.05	237/600	0.08	0.24	0.22	0.91	0.64	0.67
Mean shift	0.10	322/600	0.34	0.48	0.45	0.63	0.43	0.46
Mean shift	0.25	411/600	0.79	0.80	0.79	0.17	0.16	0.16
Mean shift	0.50	454/600	0.89	0.90	0.89	0.08	0.07	0.08
Scaling drift	0.02	241/600	0.05	0.23	0.22	0.89	0.54	0.55
Scaling drift	0.05	313/600	0.35	0.47	0.45	0.57	0.34	0.36
Scaling drift	0.10	350/600	0.56	0.66	0.65	0.34	0.22	0.22
Scaling drift	0.25	431/600	0.76	0.79	0.77	0.16	0.14	0.16
Scaling drift	0.50	485/600	0.88	0.88	0.86	0.10	0.08	0.10

TABLE 27

Gate A: detection rate of the executed drift mechanisms (matched controls) and of their adaptive cluster-preserving variants (F5), 600 observations per row pooled over the eight environments, as union / 1.2.0 rule / conformal rule in the batch-level regimes, and exact item-aligned detection (prediction disagreement or confusion-profile delta) under $\mathcal{I}^{\star}_{XFY}$.

Mechanism	Strength	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{XFY}$	$\mathcal{I}^{\star}_{XFY}$ exact
Mean shift (executed)	0.02	1.00 / 1.00 / 1.00	1.00 / 1.00 / 0.97	1.00 / 0.99 / 0.97	0.36
Mean shift (executed)	0.05	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	0.50
Mean shift (executed)	0.10	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	0.62
Mean shift (F5)	0.02	0.01 / 0.01 / 0.01	0.10 / 0.06 / 0.06	0.10 / 0.06 / 0.06	0.32
Mean shift (F5)	0.05	0.10 / 0.08 / 0.08	0.31 / 0.26 / 0.24	0.31 / 0.24 / 0.22	0.45
Mean shift (F5)	0.10	0.40 / 0.34 / 0.34	0.60 / 0.52 / 0.48	0.60 / 0.50 / 0.45	0.60
Mean shift (F5)	0.25	0.82 / 0.79 / 0.79	0.87 / 0.82 / 0.80	0.87 / 0.81 / 0.79	0.74
Mean shift (F5)	0.50	0.91 / 0.89 / 0.89	0.94 / 0.91 / 0.90	0.94 / 0.91 / 0.89	0.79
Scaling drift (executed)	0.02	1.00 / 0.99 / 0.99	1.00 / 0.99 / 0.96	1.00 / 0.98 / 0.95	0.54
Scaling drift (executed)	0.05	1.00 / 1.00 / 1.00	1.00 / 1.00 / 0.99	1.00 / 1.00 / 0.99	0.61
Scaling drift (executed)	0.10	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00	0.67
Scaling drift (F5)	0.02	0.07 / 0.05 / 0.05	0.28 / 0.24 / 0.23	0.28 / 0.23 / 0.22	0.47
Scaling drift (F5)	0.05	0.39 / 0.35 / 0.35	0.57 / 0.50 / 0.47	0.57 / 0.47 / 0.45	0.57
Scaling drift (F5)	0.10	0.61 / 0.56 / 0.56	0.71 / 0.67 / 0.66	0.71 / 0.66 / 0.65	0.64
Scaling drift (F5)	0.25	0.79 / 0.76 / 0.76	0.85 / 0.81 / 0.79	0.85 / 0.80 / 0.77	0.77
Scaling drift (F5)	0.50	0.91 / 0.88 / 0.88	0.95 / 0.90 / 0.88	0.95 / 0.89 / 0.86	0.87

TABLE 28
Gate A: prediction-change rate (item-aligned disagreement > 0), material fraction at $|\Delta_R| > 0$, ≥ 0.02 and ≥ 0.05, mean $|\Delta_R|$, fraction of rows whose balanced accuracy rose, and mean fraction of feature entries the attacker perturbs (the executed mechanisms perturb every entry).

Mechanism	Strength	Pred. change	Mat. 0^+	Mat. 0.02	Mat. 0.05	Mean $\lvert\Delta_R\rvert$	Raised	Perturbed
Mean shift (executed)	0.02	0.36	0.33	0.10	0.06	0.011	0.13	–
Mean shift (executed)	0.05	0.50	0.46	0.20	0.11	0.026	0.20	–
Mean shift (executed)	0.10	0.62	0.59	0.35	0.24	0.054	0.20	–
Mean shift (F5)	0.02	0.32	0.29	0.03	0.00	0.003	0.14	0.33
Mean shift (F5)	0.05	0.45	0.40	0.10	0.01	0.006	0.20	0.33
Mean shift (F5)	0.10	0.60	0.54	0.22	0.05	0.014	0.22	0.33
Mean shift (F5)	0.25	0.74	0.69	0.42	0.22	0.033	0.25	0.33
Mean shift (F5)	0.50	0.79	0.76	0.52	0.35	0.053	0.18	0.33
Scaling drift (executed)	0.02	0.54	0.48	0.23	0.14	0.032	0.17	–
Scaling drift (executed)	0.05	0.61	0.57	0.42	0.30	0.063	0.17	–
Scaling drift (executed)	0.10	0.67	0.64	0.50	0.41	0.096	0.19	–
Scaling drift (F5)	0.02	0.47	0.40	0.11	0.02	0.008	0.15	0.33
Scaling drift (F5)	0.05	0.57	0.52	0.27	0.11	0.019	0.16	0.33
Scaling drift (F5)	0.10	0.64	0.58	0.36	0.21	0.030	0.18	0.33
Scaling drift (F5)	0.25	0.77	0.72	0.47	0.33	0.048	0.19	0.33
Scaling drift (F5)	0.50	0.87	0.81	0.53	0.35	0.060	0.18	0.33

TABLE 30
Gate A: per-environment detection rate of the adaptive variants under the conformal rule in $\mathcal{I}_{XF}$ (rows pooled over models, dimensions, splits and seeds), and mean fraction of entries the attacker perturbs in that environment.

		Mean shift					Scaling drift				
Environment	Perturbed	0.02	0.05	0.10	0.25	0.50	0.02	0.05	0.10	0.25	0.50
E1 CICIDS ID 256	0.40	0.03	0.43	0.90	1.00	1.00	0.50	0.67	1.00	1.00	1.00
E2 UNSW-NB15 ID	0.45	0.13	0.33	0.48	0.93	0.99	0.24	0.50	0.77	0.88	0.99
E3 ToN-IoT ID	0.23	0.03	0.24	0.47	0.77	0.92	0.20	0.42	0.57	0.75	0.88
E4 CICIDS Tue→Wed	0.40	0.00	0.00	0.00	0.25	0.55	0.00	0.82	1.00	1.00	1.00
E5 CICIDS Tue→Fri-PortScan	0.19	0.00	0.02	0.32	0.65	0.75	0.00	0.12	0.25	0.43	0.60
E6 CICIDS Wed→Thu-Web	0.30	0.05	0.27	0.75	0.98	1.00	0.37	0.53	0.82	1.00	1.00
E7 CICIDS Wed→Fri-AM	0.16	0.00	0.08	0.37	0.95	0.88	0.08	0.18	0.30	0.50	0.60
E8 UNSW temporal OOD	0.45	0.18	0.43	0.60	0.78	1.00	0.42	0.50	0.57	0.70	0.90